

# Neutron Generation from Z-Pinches

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**Abstract**—Recent advances in both experimental and theoretical studies on neutron generation in various Z-pinch facilities are reviewed. The main methods for enhancing neutron emission from the Z-pinch plasma are described, and the problems of igniting a thermonuclear burn wave in this plasma are discussed.

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## 1. INTRODUCTION

Studies on discharges pinching under the action of the magnetic self-field began in the early 1950s. Artsimovich et al. [1, 2] were the first to observe the generation of thermonuclear neutrons in experiments with straight discharge tubes filled with deuterium.

The original idea of creating a neutron source and, then, a thermonuclear reactor based on a Z-pinch was as follows. A high-current discharge is initiated in a chamber filled with deuterium and tritium, and the gas-kinetic pressure of the discharge plasma is counterbalanced by the magnetic field generated by the discharge current. The radial equilibrium between the plasma and magnetic pressures implies that the temperature of the hydrogen Z-pinch plasma satisfies the Bennet condition,

$$T = I^2/4c^2N, \quad (1.1)$$

where  $N = \pi r^2 n$  is the number of ions per unit length of the pinch.

It follows from Eq. (1.1) that, for a current of  $10^6$  A and  $N = 10^{17}$  cm<sup>-3</sup>, the equilibrium temperature of the discharge plasma is 10 keV, which is sufficient to ignite a self-sustained thermonuclear reaction in the pinch plasma.

Indeed, the first experiments with Z-pinches revealed neutron emission from a deuterium plasma ( $10^8$ – $10^9$  neutrons per shot [3, 5]). These results motivated investigation of Z-pinches; however, a further, more thorough study of Z-pinch discharges showed that the above simple model fails to adequately describe the process of neutron emission from a Z-pinch [6].

The pinch turned out to be unstable against MHD instabilities [7, 8] and rapidly disrupted. In addition, it was found that neutrons were mostly generated in the phase of a highly developed instability. Therefore, the approach to pinch discharges changed radically and further investigations were focused on the behavior of an MHD unstable pinch plasma.

To date, many different Z-pinch configurations have been investigated, experiments with various deuterium-containing loads have been carried out, and the pinch current has been substantially increased. As a result, the neutron yield from pinches produced of deuterium-containing substances has been increased by five orders of the magnitude [9].

Along with this line of research, other methods of employing Z-pinches to initiate thermonuclear reaction were also developed. The main method is based on a pulsed high-power source of X radiation emitted from a Z-pinch [10, 11]. This radiation should compress a spherical thermonuclear target to ignite a self-sustained thermonuclear reaction in it. However, the review of these studies goes beyond the scope of the present paper.

## 2. REVIEW OF EXPERIMENTAL STUDIES ON NEUTRON EMISSION FROM PINCHES

### 2.1. Neutron Yield from Gas Discharges and Plasma Focuses

A considerable increase in the neutron yield (up to  $6 \times 10^{12}$  neutrons per shot) was achieved in noncylindrical Z-pinch discharges, usually referred to as plasma focuses (PFs) [12]. In PFs, the current sheath has the form of a funnel through which the plasma can escape from the region of the maximum density. It was found experimentally that neutron generation from Z-pinches is related to the development of sausage instability ( $m = 0$ ) [13, 14]. The most important feature of this instability is an explosive increase in the energy density in the nonlinear stage of the constriction development. In studying how the plasma is accelerated by the discharge current in coaxial plasma accelerators, Mather discovered the effect of axial plasma focusing near the anode [15]. By analogy with the focusing of light beams, the place where the plasma was focused was called the PF. In many respects, the effect of axial

plasma focusing in PF facilities resembles the effect of plasma implosion in noncylindrical Z-pinches.

Even the early experiments demonstrated that the neutron yield depended strongly on the discharge current. Important experimental data on the mechanisms for this effect was obtained in studying the ion and electron emission, hard X-ray (HXR) and soft X-ray (SXR) emission, and the dynamics of the formation and compression of a hot dense plasma. It was established that the neutron yield  $Y$  increased with discharge current  $I$  according to a power law  $Y \propto I^n$ , where  $n$  is in the range 3.3–5 [16–21]. Filippov and Imshennik [16] found that the neutron yield from an optimized PF was described by the scaling  $Y \propto I^{4.9}$ . The dependence of the total neutron yield on the discharge current in the POSEIDON PF facility ( $W = 680$  kJ,  $U = 60$  kV) with a current of up to 1 MA was found to obey the power law  $Y \propto I^4$  [18], whereas in experiments [19, 20] carried out in other PF facilities, the scaling  $Y \propto I^{4.4}$  was observed. An analysis of a large amount of experimental data showed that, in most of experiments, the scaling exponent was close to 3.3 [21]. It should be noted, however, that the above dependences of the neutron yield on the discharge current were observed up to currents of 2 MA; for higher currents, the neutron yield saturated.

A large number of works were devoted to investigations of neutron radiation from PF discharges [18, 19, 22–28]. The main parameters of these facilities are given in Table 1. It was found that the energy spectrum of neutrons in Mather-type facilities is rather complicated and has several maxima. In experiments carried out in the Frascati megajoule facility ( $W = 0.5$  MJ,  $U = 28$  kV,  $I = 0.8$  MA, and  $Y = 10^{11}$  neutrons per shot), the energy spectrum was measured by the time-of-flight technique with the use of scintillation detectors oriented at the right angle to the axis of the discharge chamber. In most of these experiments, a central peak at the energy 2.45 MeV, corresponding to thermal generation of neutrons, and two side peaks were observed [26]. The authors attributed these peaks to the generation of deuteron beams of complicated configuration. The results of measurements carried out in [23, 24] by the method of nuclear emulsions showed that the position and shape of the spectral maximum depended on the observation angle  $\theta$  ( $\theta = 0^\circ$  corresponded to observations along the pinch axis, from the anode to the cathode). In those experiments, the maximum of the energy spectrum was found to shift from 2.45 MeV ( $\theta = 90^\circ$ ) toward higher energies with decreasing observation angle. This effect was observed at angles  $\theta \leq 45^\circ$  (see Fig. 1). The shift was the largest at  $\theta = 0^\circ$ , the peak of the spectrum being at 2.85 MeV. Moreover, additional peaks with lower amplitudes arose, and the spectrum became asymmetric. Time-of-flight measurements of the neutron energy in the axial direction in the PF-1000 facility ( $W = 1$  MJ,  $U = 40$  kV,  $I = 2.3$  MA, and  $Y = 10^{10}$ – $10^{11}$  neutrons per shot) showed that the neutron energy observed at the angle  $\theta = 0^\circ$  was shifted from

2.45 MeV toward higher energies and was equal to 2.7 MeV, whereas at  $\theta = 180^\circ$ , it was shifted toward lower energies and was equal to 2.12 MeV [27].

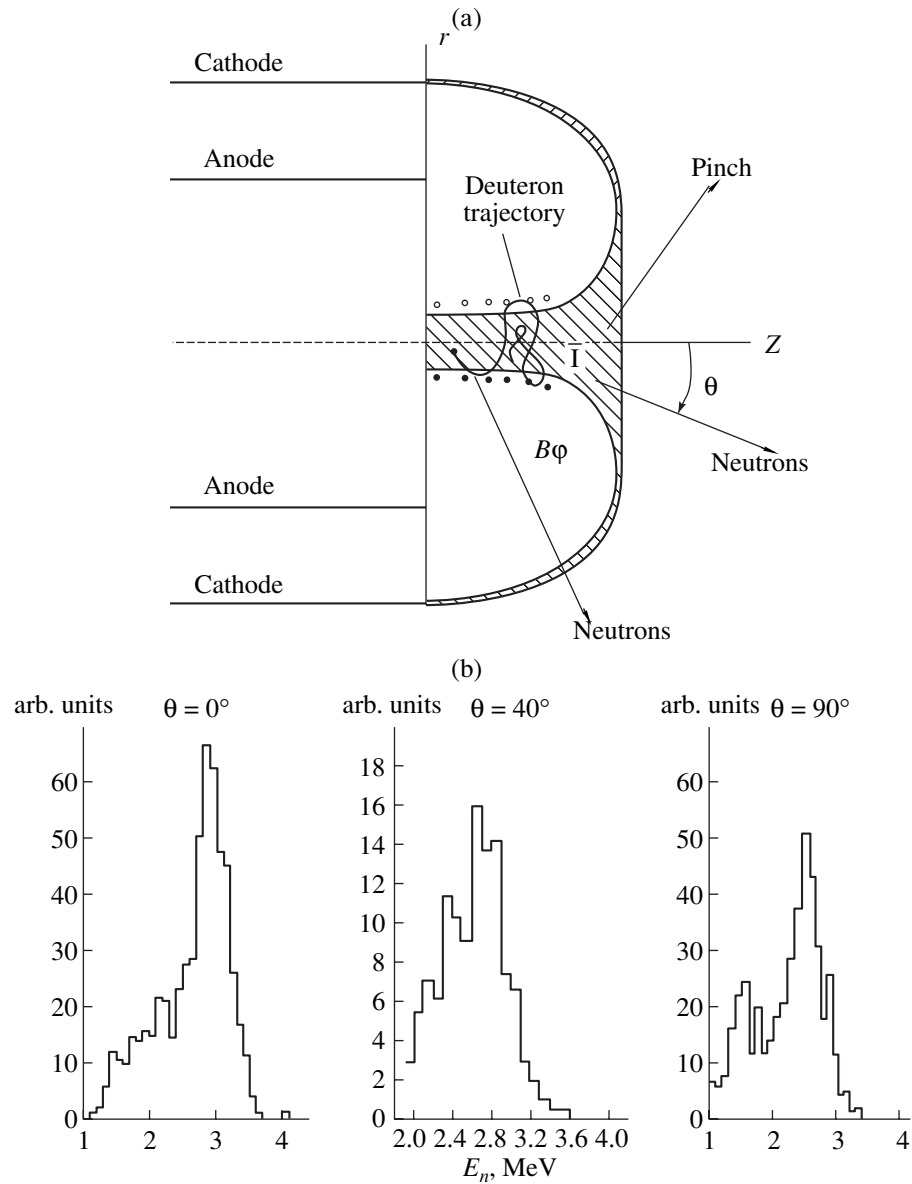
Time-of-flight measurements performed in [22] showed that the neutron energy at the leading edge of the two recorded neutron pulses observed at the angle  $\theta = 0^\circ$  extended to 2.8–3.2 MeV, whereas the energy of the last group of neutrons corresponding to the trailing edge extended to 2.6–2.7 MeV. The spatial distribution of neutron emission in the last group of neutrons was nearly isotropic. The emergence of the first and second pulses of neutron emission correlated with the first and second compressions of the plasma current sheath, respectively, and with the formation of a spherical plasma structure after each compression of the current sheath.

In attempting to construct a consistent physical model of neutron generation, it was supposed that the above results may be attributed to the generation of local deuteron beams, which are then accelerated in the axial and azimuthal directions to an energy of  $\sim 100$  keV.

In experiments carried out in PF facilities, much attention was paid to studying the spatiotemporal distribution of neutron emission. The anisotropy of the total neutron yield was measured in [21, 24, 28]. In the PF-360 facility ( $W = 360$  kJ,  $U = 50$  kV,  $I = 1.85$  MA, and  $Y = 10^{10}$  neutrons per shot), as well as in experiments carried out in similar facilities, the neutron emission anisotropy, defined as the ratio  $Y(\theta)/Y(\theta = 90^\circ)$ , was in the range 1.7–2.0 (Fig. 2). In the PF-360 experiments, the neutron yield was maximum along the pinch axis and was minimum at angles of  $100^\circ$ – $160^\circ$  with respect to the axis.

The temporal structure of neutron pulses was investigated in [18, 19, 27, 28]. Two or three neutron pulses were usually observed in one shot. The first pulse, with a full width at half-maximum (FWHM) of 50 ns, contributed only 10–20% to the total neutron yield, whereas the main contribution (80–90%) came from the subsequent pulse, recorded  $\approx 150$  ns after the first one. The time dependences of the neutron flux  $dY/dt$  and the FWHMs of the neutron energy spectrum  $\Delta E_n$  measured in the POSEIDON facility ( $I = 1$  MA,  $Y = 10^{10}$ – $10^{11}$  neutrons per shot) are shown in Fig. 3a. The neutron energy spectrum in two successive pulses is presented in Fig. 3b. Space- and time-resolved measurements performed in [17, 27] showed that, in the second pulse, the neutron emission anisotropy was  $Y(\theta = 0^\circ)/Y(\theta = 90^\circ) = 1.82$ , whereas in the first pulse, it was considerably smaller (about 1.2).

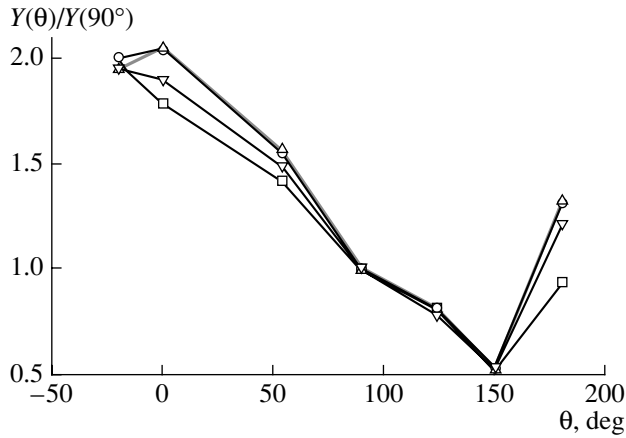
The dimensions, shape, and position of the neutron generation region were measured in [29, 30] using a collimated scintillation photomultiplier as a neutron detector. Measurements performed in a Filippov-type PF facility ( $W = 120$  kJ,  $U = 40$  kV, and  $Y = 10^{10}$  neutrons per shot) [30] showed that 60–70% of neutrons were emitted 100 ns after the maximum compression from a 3-cm-diameter plasma column ( $n_e = 3 \times$



**Fig. 1.** (a) Schematic of a Mather-type PF facility; (b) neutron energy spectra as functions of the observation angle  $\theta$ .

$10^{17} \text{ cm}^{-3}$ ,  $T_i = 7\text{--}8 \text{ keV}$ ) located at a distance of 1–3 cm from the anode, 15–20% were emitted 120 ns after the maximum compression from a narrow zone located at a distance of 6 cm from the anode (near the cathode), and 10–15% were emitted 140 ns after the maximum compression from the entire plasma region between the anode and cathode. In experiments carried out in a Filippov-type facility [29] with a total neutral yield less than  $5 \times 10^{10}$  neutrons per shot, an image of the neutron generation region was obtained with the use of a bubble chamber. This region was axisymmetric in shape, 8–10 cm in length, and 2 mm in diameter, its centroid being located at the axis of the discharge chamber, at a distance of 12 mm from the anode.

The deuteron energy spectra were measured in [20, 31–35]. In [31], the spectra of deuterons emitted at an angle of  $45^\circ$  with respect to the axis of a Filippov-type facility ( $W = 115 \text{ kJ}$ ,  $U = 20 \text{ kV}$ ,  $I = 1 \text{ MA}$ ), operating at a pressure of  $p = 0.4\text{--}1.2 \text{ Torr}$  (deuterium with an admixture of nitrogen and xenon), were measured by the method of nuclear emulsions. In experiments performed with a plane anode, the spectrum of fast ions ( $0.2 < E < 1.5 \text{ MeV}$ ) was observed to have several maxima [32]. The ion temperature determined from the relative rate of the DD and  $D^3\text{He}$  reactions in  $D\text{--}^3\text{He}$  discharges at pressures of up to 0.35 Torr was found to be  $T_i > 5.3 \text{ keV}$ , and that estimated from the FWHM of the energy spectrum ( $\delta E = 0.6 \text{ MeV}$ ) in the range 2.4–



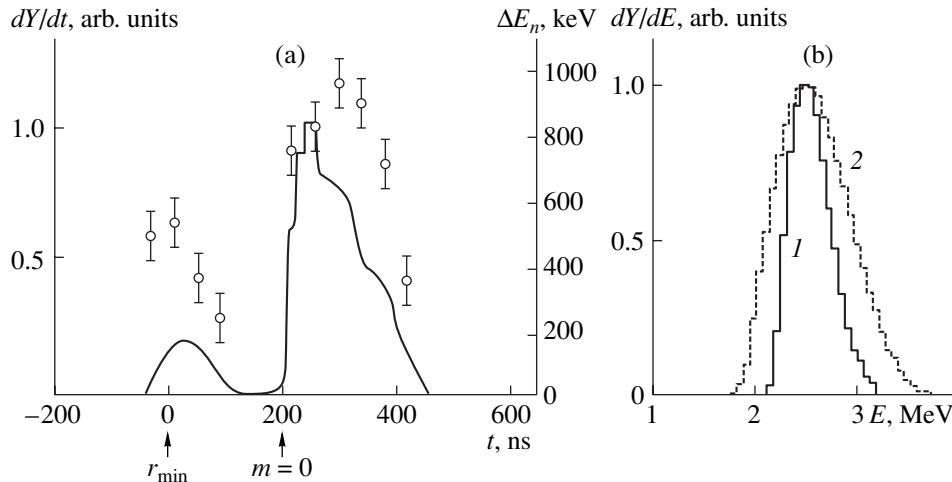
**Fig. 2.** Normalized neutron yield  $Y(\theta)/Y(90^\circ)$  vs. observation angle  $\theta$  for different deuterium pressures:  $p = 5.3$  Torr (squares), 6.2 Torr (circles), 6.8 Torr (triangles), and 7.6 Torr (inverted triangles).

3.5 MeV with a maximum near 3 MeV was found to be  $T_i < 40$  keV. The energy spectrum as a function of the gas pressure was investigated in [35]. At low pressures (0.26–1.9 Torr), the deuteron spectrum in the energy ranges  $E < 1.5$  MeV and  $E = 1.5$ –3 MeV was found to fall off exponentially with different effective temperatures. As the pressure was increased to 5 Torr, the spectrum was depleted of fast ions.

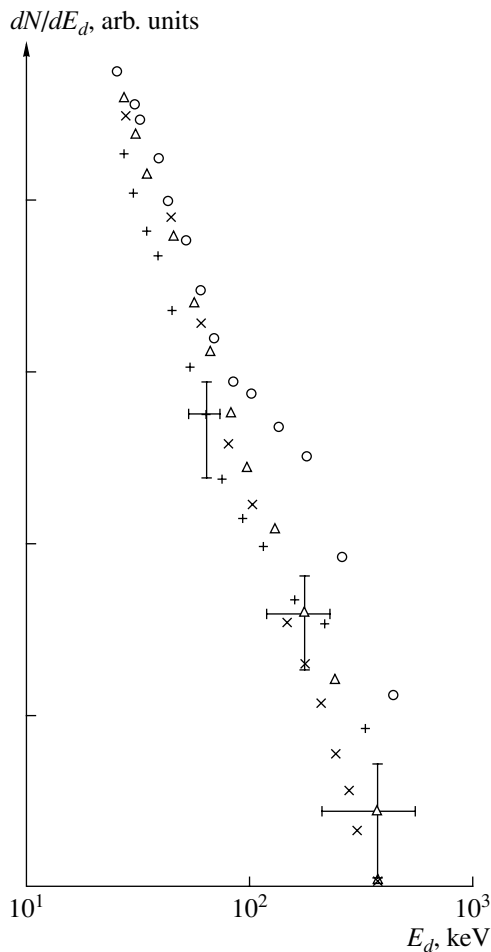
In experiments performed in a Mather-type facility ( $U = 25$  kV,  $I = 560$  kA,  $Y = 7.5 \times 10^9$  neutrons per shot) [20], operating at a deuterium pressure of  $p > 3$  Torr, both the ion and electron spectra were measured simul-

taneously. Unlike in other works, the time-of-flight technique employed in [20] made it possible to measure the ion spectrum in the range of lower energies (down to 30 keV). The spectra of charged particle flows in the energy range 30–600 keV were measured along the PF axis. The ion spectrum  $dN/dE_d$  was found to fall off according to a power law with an exponent of  $3.4 \pm 0.4$  (see Fig. 4). The electron spectrum  $dN/dE_e$  also fell off according to a power law with nearly the same exponent. It was found that the neutron yield in those experiments depended on the discharge current as  $Y \propto I^{4.4}$ . In the PF-100 facility ( $W = 60$  kJ,  $U = 20$  kV,  $I = 1$  MA) [34], the ion energy spectrum in the energy range 0.1–5 MeV was measured using a Thomson analyzer. Two groups of ions were detected in the deuteron spectrum: in the energy range of 0.15–1 MeV, the spectrum decreased as  $dN/dE \sim E^{-3}$ , whereas at higher energies, it decreased more slowly, as  $dN/dE \sim E^{-2}$ .

The angular distribution of deuterons was studied in [18, 36, 37]. In [36], intense deuteron beams with energies of  $E_d > 330$  keV were found to be emitted in the shape of narrow cathode-directed jets with a divergence angle of  $7^\circ$ , within which more than one-half of deuterons were contained, the angular distribution of the deuterons being close to exponential,  $N_d \sim \exp(\alpha\theta)$ , where  $\alpha$  is a constant factor and  $\theta$  is the angle with respect to the chamber axis. Deuteron emission in the energy range 0.07–2.5 MeV was investigated in [18]. Measurements of the angular distribution of deuterons showed that, when the detection angle  $\theta$  was increased from  $30^\circ$  to  $110^\circ$  ( $\theta$ , the number of deuterons with energies lower than 1.5 MeV became three orders of magnitude larger than the number of deuterons in the energy range 2.7–



**Fig. 3.** (a) Time dependences of the neutron flux  $dY/dt$  (solid curve) and the FWHM of the neutron energy spectrum  $\Delta E_n$  at a deuterium pressure of  $p = 6$  Torr ( $t = 0$  refers to the instant of maximum compression, and  $t = 200$  ns corresponds to the onset of the  $m = 0$  mode of instability); (b) neutron energy spectra in two subsequent neutron pulses recorded at different instants in the same shot: (1)  $t = 0$  and (2)  $t = 280$  ns (the instants correspond to maxima of the neutron intensity). The total neutron yields in the first and second pulses are  $Y = 6 \times 10^9$  and  $4.1 \times 10^{10}$ , respectively.



**Fig. 4.** Energy spectra of deuterium ions,  $dN/dE_d$ , recorded in four different shots: shot no. 1 (triangles), shot no. 2 (pluses), shot no. 3 (crosses), and shot no. 4 (circles). The experiments were performed at a deuterium pressure of 3 Torr, the neutron yield being  $(2.2 \times 10^9) \pm 10\%$ .

3.7 MeV. It was found that the deuteron energy spectrum fell off abruptly in the energy range 1–1.5 MeV and that the high-energy component of the spectrum decreased with increasing current.

The time behavior of ion emission was studied in [33, 37]. In [33], the flux of ions with energies of 0.3–3 MeV was measured at a distance of 2 m from the PF by a Thomson parabolic ion analyzer with a time resolution of  $<4$  ns. Two types of ion pulses were recorded: short solitary pulses with an FWHM of 8–10 ns and longer pulses of complicated structure. The generation of ions was accompanied by the generation of electrons and HXR emission. Collimated radiation measurements showed that the solitary ion pulses were emitted from a hot spot ( $E > 2$  keV) located near the pinch axis, whereas the longer pulses of complicated structure were emitted from several hot spots located both on and off the axis.

Thus, the studies of PFs showed that, in the energy range 30–600 keV, the spectra  $dN/dE_d$  of charged particles emitted along the PF axis fall off according to a power law with an exponent of  $-3.4 \pm 0.4$ . In the energy range 0.1–5 MeV, two groups of ions with different spectral shapes were revealed in the deuteron spectrum. In the energy range 0.15–1 MeV, the spectrum fell off as  $dN/dE \sim E^{-3}$ , whereas at energies higher than 1 MeV, it fell off more slowly, as  $dN/dE \sim E^{-2}$ . Space- and time-resolved measurements of fast ions showed that two types of ion pulses were generated in the energy range 0.3–3 MeV: short solitary pulses with an FWHM of 8–10 ns were emitted from a single hot spot, whereas longer pulses of complicated structure were emitted from several hot spots. The ions were emitted from small-size ( $<0.5$  mm) regions located near the pinch axis, the ion temperature  $T_i$  being in the range 5.3–9 keV.

The dependence of the neutron yield on the discharge current is described by a power-law function  $Y \propto I^n$ , where  $n = 3.3$ –5. Space- and time-resolved measurements of neutron radiation showed that the second neutron pulse contributed as large as 80–90% to the neutron yield, whereas the first one contributed only 10–20%. The neutron radiation anisotropy is in the range 1.7–2.0 and is appreciably higher for the second pulse than for the first one.

## 2.2. Dense Z-Pinches

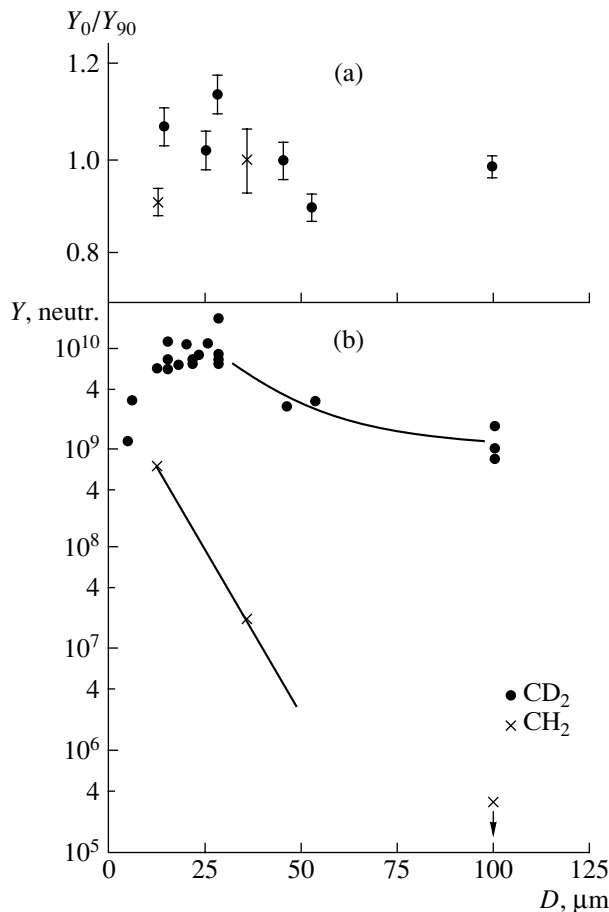
In recent years, Z-pinches are expected to be able to be compressed to very small radii and high densities due to radiative collapse [38]. Unsurpassed plasma parameters were achieved in experiments with vacuum sparks [39, 40]. At currents of 100–200 kA, the plasma temperature and density in hot spots reached 5 keV and  $10^{23}$  cm $^{-3}$ , respectively, the hot spot size being 2–10  $\mu$ m.

Recent progress in studying Z-pinches was achieved due to creating high-power ( $>50$  TW) nanosecond current generators (up to 20 MA) with a current rise rate of  $10^{14}$  A/s. This allowed one to investigate Z-pinches with a transverse size as small as  $\sim 1$  mm and an initial density close to the density of solids. A transition to hydrogen pinches made it possible to study the compression of hydrogen, whose properties are most close to those of a thermonuclear fuel. It was understood in the mid-1980s that, in order to reach a high energy density in a pinch, it is expedient that the discharge be initiated in a condensed hydrogen [41, 42]. Radiation losses from a hydrogen Z-pinch can significantly affect its parameters when the pinch current exceeds the Braginskii–Pease current,

$$I_{BP} [\text{MA}] = \frac{0.665 \frac{Z+1}{Z}}{\sqrt{\frac{\lambda \ln \Lambda}{10}}}, \quad (2.2)$$

**Table 1.** The main parameters of PF facilities

| Science center                                                     | Facility, type, reference          | Vacuum chamber; $d_{\text{int}}/d_{\text{ext}}; L, \text{ mm}$ | Load (gas); pressure, Torr | Stored energy, MJ | Discharge current, MA | $n_e, \text{ cm}^{-3}$ | $T_{\text{max}}, \text{ keV}$  | Diameter/length $d/L, \text{ cm}$ ; Pinch lifetime, ns | Neutron yield $Y$                               | Neutron energy $E, \text{ MeV}$                                                                         |
|--------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------|----------------------------|-------------------|-----------------------|------------------------|--------------------------------|--------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Frascati, Rome (Italy)                                             | Mather-type PF facility [24]       | $370 \times 300$                                               | Deuterium, 1.5–8           | 1                 | 2.8                   | $10^{19}$              | 9                              | 0.8/5<br>100                                           | $7 \times 10^{11}$                              | 2.5–3.0                                                                                                 |
| Institute of Nuclear physics, Warsaw (Poland)                      | PF-360 Mather-type facility [21]   | 100/150 $\times$ 300                                           | Deuterium, 4–10            | 0.18              | 1.85                  | $10^{19}$              |                                | 0.5/2<br>110–120                                       | $3.8 \times 10^{10}$<br>$Y \propto I^{3.2-3.3}$ | 2.5                                                                                                     |
| Institute of Plasma Research, Stuttgart (Germany)                  | POSEIDON Mather-type facility [18] | 208/270 $\times$ 470                                           | Deuterium, 2–6             | 0.5               | 1.5                   | $10^{19}$              | $T_i \leq 1.5$<br>$T_e = 1$    | 2/7<br>100–130                                         | $2 \times 10^{11}$<br>$Y \propto I^{3.2-3.3}$   | 2.1–3.2 (1 imp)<br>1.8–3.5 (2 imp)                                                                      |
| Kurchatov Institute, Moscow (Russia)                               | PF-3 Filippov-type facility [13]   | 1000/2500 $\times$ 500                                         | Deuterium, 0.5–10          | 2.8               | 3                     | $10^{19}$ – $10^{20}$  | $T_i = 5$                      | 0.5–1.5/10<br>200                                      | $10^{11}$<br>$Y \propto I^4$                    | 2.5                                                                                                     |
| Institute of Plasma Physics and Laser Microfusion, Warsaw (Poland) | PF-1000 Mather-type facility [27]  | 232/400 $\times$ 600                                           | Deuterium, 4–8             | 1                 | 2.3                   | $5 \times 10^{19}$     | $T_i \approx 2$<br>$T_e = 0.6$ | 0.5/5<br>120                                           | $3 \times 10^{11}$                              | $E_{\text{max}} = 3.2$<br>( $\theta = 0^\circ$ )<br>$E_{\text{max}} = 2.7$<br>( $\theta = 90^\circ$ )   |
| Institute of Experimental Physics, Sarov (Russia)                  | Filippov-type facility [29]        |                                                                | D–T mixture, 18–23         | 0.12              | 1.2                   |                        | $T_i = 3$ –5                   | 0.2/0.5–1.2<br>20 (neutron source parameters)          | $1.5 \times 10^{11}$                            | $E_{\text{max}} = 14.7$<br>( $\theta = 0^\circ$ )<br>$E_{\text{max}} = 13.5$<br>( $\theta = 90^\circ$ ) |



**Fig. 5.** (a) Anisotropy of the neutron yield,  $Y_0/Y_{90}$  (where  $Y_0$  and  $Y_{90}$  are the integral neutron fluxes measured along the pinch axis and at the right angle to it); (b) total neutron yield  $Y$  as a function of the diameter of deuterated ( $\text{CD}_2$ ) and polyethylene ( $\text{CH}_2$ ) fibers.

where  $Z$  is the mean ion charge;  $\ln \Lambda$  is the Coulomb logarithm, which can vary from 5 to 15 in different experiments;  $\lambda$  is a factor depending on the distributions of the plasma density, electron temperature, and current density and is close to unity in most cases; and  $I_{\text{BP}} \approx 1.4$  MA for hydrogen. Radiation compression of fine (30–120  $\mu\text{m}$  in diameter) long (1–4 cm) fibers at currents of up to 1.2 MA was studied experimentally in many laboratories [43–46]. A necessary requirement for such experiments is stable radiative compression over many inertial times  $\tau_i = r/v$ , where  $r$  is the pinch radius and  $v$  is the plasma velocity. The parameters of some facilities are given in Table 2.

In the POSEIDON ( $U = 600$  kV,  $I = 640$  kA,  $\tau = 130$  ns) and ZFX ( $I = 1.8$  MA,  $U = 840$  kV,  $\tau = 610$  ns) facilities, an enhanced stability of the plasma channel formed in an electric discharge through a frozen fiber was observed during the first (stable) confinement phase, lasting over 10–100 Alfvén times [43, 44]. However, the development of the  $m = 0$  mode of hydrody-

namic instability in the next (so-called dynamic) phase led to the separation of the plasma channel into individual filaments with a temperature as high as 0.4–1 keV. This impeded the further stable compression of the entire channel. An analysis of experimental data obtained at a shorter current rise time ( $\tau_{\text{fr}} = 150$  ns) in the MAGPIE facility ( $U = 2.4$  MV,  $I = 1.4$  MA) led to the conclusion that the time behavior of 80- $\mu\text{m}$ -diameter frozen hydrogen fibers was similar to that of graphite and frozen deuterium fibers [46]. Simulations of the compression dynamics of 33- to 300- $\mu\text{m}$ -diameter graphite and frozen deuterium fibers showed that the latter were unstable against the  $m = 0$  mode of hydrodynamic instability.

**2.2.1. Experiments with deuterated fibers.** The first experiments aimed at generating thermonuclear neutrons from deuterated fibers were carried out in [47–49]. The neutron yield achieved in the Don facility, in which the current was passed through 10- $\mu\text{m}$  deuterated polyethylene fibers, was  $10^6$ – $10^7$ . As the fiber diameter was increased to 30  $\mu\text{m}$ , the neutron yield fell abruptly [49]. The 80-ns neutron pulse consisted of several short spikes, which were attributed by the authors to the generation of neutrons in individual constrictions arising at different instants.

A fairly high neutron yield ( $Y = 10^9$ – $10^{10}$ ) was observed in the Gamble-II facility ( $\rho = 1.5 \Omega$ ,  $U = 1$  MV,  $\tau = 50$  ns) [48], in which current pulses with an amplitude of up to 0.7 MA and duration of 50 ns were passed through deuterated fibers ( $\text{CD}_2$ ) 8–400  $\mu\text{m}$  in diameter and 3 cm in length (see Fig. 5). In experiments with fibers less than 50  $\mu\text{m}$  in diameter, neutrons with energies of several megaelectronvolts were detected. The neutrons could be generated in the  $d(d, n)^3\text{He}$  and  $^{12}\text{C}(d, n)^{13}\text{N}$  reactions, initiated by accelerated deuterons, whose number exceeded  $10^{13}$ . When the diameter of fibers was increased to more than 50  $\mu\text{m}$ , neutrons with energies of about 2.45 MeV were detected, whose spectrum corresponded to a thermal distribution of the colliding ions. The neutron emission anisotropy (the ratio of the neutron yield measured along the pinch to that measured at the right angle to the pinch axis) was about 0.9–1.2. The observed decrease in the neutron yield with increasing fiber diameter was caused by a change in the mechanism for neutron generation, as well as by a decrease in the efficiency of energy transfer from the generator to the plasma generated in the electric explosion of a fiber. The measurements showed that the energy transferred from the generator to the plasma did not exceed 10 kJ for fibers with diameters  $>50 \mu\text{m}$  and reached 30–35 kJ for fibers with diameters  $<25 \mu\text{m}$ .

An analysis of the experimental data obtained with fibers  $>50 \mu\text{m}$  in diameter showed that the neutrons were generated in the D–D fusion reaction. For a fiber diameter of 50  $\mu\text{m}$ , the neutron yield corresponded to the temperature  $T_i = 1$  keV and  $n\tau \approx 10^{13} \text{ cm}^{-3} \text{ s}$ , whereas for fibers with diameters  $<50 \mu\text{m}$ , the neutron emission was caused by the generation of fast ions. The

**Table 2.** The main parameters of high-current generators

| Science center                                                               | Facility, type, reference    | Energy,<br>$W_{\text{gen}}/W_{\text{load}}$ ,<br>kJ | Load, diameter/length,<br>mm/cm                                            | $I_{\text{max}}/U_{\text{max}}$ ,<br>MA/MV;<br>pulse duration, ns | $n_e$ , $\text{cm}^{-3}$       | $T_{\text{max}}$ ,<br>keV            | Pinch diameter/<br>Pinch lifetime,<br>mm/ns | Neutron yield $Y$                     | Neutron energy<br>$E$ , MeV                              |
|------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------|--------------------------------------|---------------------------------------------|---------------------------------------|----------------------------------------------------------|
| Naval Research Laboratory,<br>Washington (USA)                               | Famble-2 generator [48]      | 50/10–35                                            | CD <sub>2</sub> fiber<br>0.008–0.4/3                                       | 0.7/1<br>50                                                       |                                | 1                                    | $\leq 1/100$                                | $10^{10}$                             | $\leq 10$ ( $d < 0.025$ )<br>2.5 ( $d > 0.05$ )          |
| Imperial College, London (UK)                                                | MAGPIE generator [46]        | 336/215                                             | CD <sub>2</sub> fiber<br>0.05–0.2                                          | 1.4/2.4<br>200                                                    |                                | 1 (GT)                               | $\sim 0.1/40$<br>(GT)                       | $9 \times 10^8$<br>$Y \propto I^5$    | $E_{\text{max}} = 3.3\text{--}4.8$                       |
| Institute of Plasma Research,<br>Stuttgart (Germany)                         | KALIF generator [56]         | 86/50                                               | CD <sub>2</sub> fiber<br>0.015–0.15/3–5                                    | 0.8/1.8<br>60                                                     | $10^{19} \leq 10^{22}$<br>(GT) | $T_e \sim 1$<br>$T_e \leq 2$<br>(GT) | $\sim 1/50$<br>$0.05/\sim 1$<br>(GT)        | $10^{10}$                             |                                                          |
| Naval Research Laboratory,<br>Washington (USA)                               | POSEIDON generator<br>[43]   | 30                                                  | Frozen deuterium fiber                                                     | 0.64/0.6                                                          |                                |                                      | 0.34/130                                    | $8.4 \times 10^9$                     | 2.45                                                     |
| Naval Research Laboratory,<br>Washington (USA)                               | ZFX generator [44]           | 350                                                 | 0.08–0.125/5<br>Frozen deuterium fiber                                     | 130<br>2/0.75                                                     | $6 \times 10^{22}$             |                                      | $\leq 1/150$                                | $Y \propto I^{10}$<br>$4 \times 10^9$ | 2.45                                                     |
| Los Alamos National Laboratory,<br>Los Alamos (USA)                          | HDZP-2 generator [45]        | 200                                                 | 0.04–0.125/3.8<br>Frozen deuterium fiber                                   | 670<br>1.2/2                                                      | $2 \times 10^{21}$             | $> 10$                               | 0.4/100                                     | $Y \propto I^5$<br>$8 \times 10^9$    | 2.45                                                     |
| Kurchatov Institute, Moscow<br>(Russia)                                      | S-300 generator [67]         | 250/50                                              | 0.03/5<br>Agar-agar fiber                                                  | 100<br>3/0.7                                                      | $10^{22}$                      | $T_i = 12$<br>$T_e = 2$              | 0.04/5                                      | $2 \times 10^8$                       | 2.45                                                     |
| Troitsk Institute for Innovation<br>and Fusion Research, Troitsk<br>(Russia) | Angara-5-1 generator<br>[68] | 650/140                                             | 3–5/1<br>Deuterium jet                                                     | 100<br>5/1.2                                                      | $10^{20}$                      | 1                                    | 1/15                                        | $10^{12}$                             | 1.8–2.0 ( $\theta = 0$ )<br>2.45 ( $\theta = 90^\circ$ ) |
| Sandia National Laboratories,<br>Albuquerque (USA)                           | Z generator [9, 58]          | 11 400/4000                                         | 15–30/2–4<br>Two hollow deuterium<br>jets, $d_1 = 20$ mm,<br>$d_2 = 40$ mm | 150<br>20/2<br>100                                                |                                |                                      | 10–15<br>(length)                           | $6 \times 10^{13}$                    |                                                          |



deuteron energy measured by the activation technique and the nuclear emulsion method extended to 10 MeV. The long neutron pulse with neutron energies of a few megaelectronvolts also indicated the presence of ions with very high energies. It was found that the neutrons were generated in the interaction of fast deuterons with the chamber wall, the anode, and the cathode, as well as in reactions involving other ions.

The emergence of fast ions in the Gamble-II facility was attributed to the development of the  $m = 0$  mode of hydrodynamic instability. Deuteron fluxes were detected in the direction perpendicular to the pinch axis. The fast ions could be generated in regions adjacent to the pinch axis and then be accelerated perpendicularly to it [50] due to the onset of sausage instability. The deuteron fluxes generated along the pinch axis were found to be asymmetric: in accordance with theoretical predictions [51], the number of ions detected on the anode side was one order of magnitude larger than that detected on the cathode side. The efficient energy transfer from the current generator to the pinch plasma can also be associated with the onset of the sausage instability, which can lead to an anomalously high plasma resistance [52, 53].

The parameters of neutron emission as functions of the electrode polarity and the diameter (50–200  $\mu\text{m}$ ) of  $\text{CD}_2$  fibers were investigated in the MAGPIE facility ( $U = 2.4$  MV,  $I = 1.4$  MA,  $\tau = 150$  ns) [54, 55]. The maximum neutron yield ( $7 \times 10^8$  neutrons per shot) was observed in experiments with 90- $\mu\text{m}$  fibers. A change in the electrode polarity was also found to affect the parameters of neutron emission. The average anisotropy of the neutron flux was  $Y(0^\circ)/Y(90^\circ) \approx 1.2$ , where  $Y(0^\circ)$  is the neutron yield recorded behind the anode and  $Y(90^\circ)$  is that measured at the right angle to the axis. When the polarity of electrodes was changed, the anisotropy increased to  $Y(180^\circ)/Y(90^\circ) \approx 1.7$ , where  $Y(180^\circ)$  is the neutron yield measured behind the cathode. Time-of-flight measurements showed that the maximum energy of neutrons emitted at an angle of  $45^\circ$  with respect to the pinch axis was 4.8 MeV, whereas the maximum energy of neutrons emitted at an angle of  $90^\circ$  was 3.3 MeV. When the electrode polarity was changed, the energy anisotropy increased from 1.5 to 1.7. An analysis of these results showed that the observed neutron emission anisotropy could be attributed to the generation of neutrons by a 2.6-MeV deuteron beam. This agrees with the electron energy estimated from measurements of HXR emission. Unlike earlier experiments carried out in the MAGPIE facility, the neutron yield depended strongly on the pinch current,  $Y \propto I^5$  (for fibers of diameter 125  $\mu\text{m}$ ). It was also found that the deuterons were accelerated in both directions along the pinch axis: beams of accelerated deuterons were recorded both behind the anode and the cathode.

The high neutron yield achieved in the Gamble-II accelerator was reproduced in experiments with fine (50–150  $\mu\text{m}$  in diameter) long ( $l = 30$ –50 mm) deuterated ( $\text{CD}_2$ ) fibers in the KALIF facility ( $U = 1.8$  MV,  $I = 0.8$  MA,  $\tau = 60$  ns) [56]. It was found that the total neutron yield depended on the presence of a specially produced prepulse ( $U_{\text{fp}} > 200$  kV,  $I = 50$  kA,  $\tau = 20$  ns) preceding the main pulse by  $\Delta t = 100$  ns. In discharges with a prepulse, the neutron yield was higher than that without such a pulse. An analysis of electrotechnical measurements showed that this was related to the fact that the current density near the discharge axis was higher in the presence of a prepulse. In both cases, the plasma column consisted of the central glowing core a few mm in diameter and of a slightly glowing, slowly expanding halo  $\approx 10$  mm in diameter. In experiments with a prepulse, the higher current density at the pinch axis resulted in the more efficient generation of hot ( $\sim 1$ -keV) plasma spots ( $\leq 50$   $\mu\text{m}$  in size), which were located along the pinch axis and could be responsible for neutron generation.

### 2.2.2. Experiments with frozen deuterium fibers.

In experiments carried out in the POSEIDON facility ( $U = 400$ –600 kV) [43], current pulses with an amplitude of 640 MA and a rise time of 130 ns were passed through 80- to 100- $\mu\text{m}$ -diameter frozen deuterium fibers. In those experiments, neutrons with energies of 2.45 MeV were detected. The neutron yield in the pinch current range 300–640 kA was found to depend strongly on the current:  $Y \propto I^{10}$ , the total neutron yield reaching  $4 \times 10^9$  neutrons per shot. A solitary neutron pulse arose at the peak of the current, when the  $m = 0$  mode of instability developed at the pinch axis. The compression of the pinch plasma in the constrictions was accompanied by the rapid plasma expansion in the adjacent regions. In [43], most of neutron emission was attributed to the formation of hot dense plasma regions on the pinch axis.

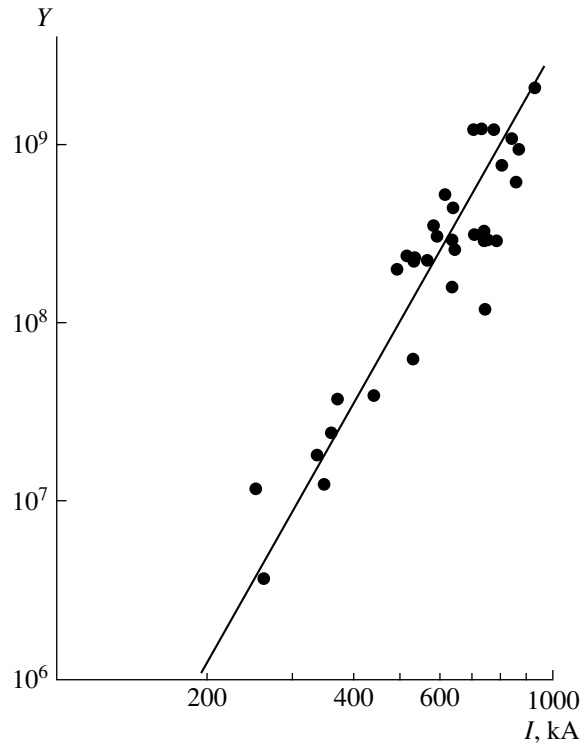
Experiments in the ZFX facility ( $U = 750$  kV,  $I = 1.8$  MA,  $\tau = 610$  ns) were also carried out at a lower current rise rate,  $dI/dt = 1.5$  kA/ns (the earlier experiments were performed at  $dI/dt = 4$ –5 kA/ns). In those experiments, current pulses with an amplitude of 920 kA and a duration of  $\tau = 840$  ns were passed through 125- $\mu\text{m}$  frozen deuterium fibers. It was found that, in this case, the pinch also evolved in two phases: the stable phase with a duration of 10 Alfvén times (in the earlier experiments carried out in the POSEIDON facility, this phase lasted up to 100 Alfvén times) and the dynamic phase, in which  $m = 0$  modes developed [44] and the pinch was divided into spatially separated regions located along the axis. As the pinch expanded with a velocity of  $0.4 \times 10^6$  cm/s during a few tens of nanoseconds, its radius increased by a factor of 4–6, but its topology remained unchanged over 250–400 ns. As in the earlier experiments, neutron emission appeared in the dynamic phase. In this phase, the plasma channel divided into spatially separated hot regions with a tem-

perature of 0.4–1 keV due to the development of the  $m = 0$  mode of hydrodynamic instability. The onset of instability was accompanied by the generation of X-ray and neutron emission (with a maximum yield of  $4 \times 10^9$  neutrons per shot). Measurements of the neutron emission anisotropy showed that the center-of-mass velocity of the interacting deuterons was directed toward the cathode. For 125- $\mu\text{m}$  frozen deuterium fibers, the neutron yield was found to depend on the current as  $Y \propto I^5$  (see Fig. 6).

A considerable neutron yield ( $5 \times 10^9$  neutrons per shot) was observed in the HDZP-II facility ( $U = 2$  MV,  $I = 1.2$  MA,  $\tau = 100$  ns) when current pulses with an amplitude of 600–800 kA were passed through frozen deuterium fibers 30  $\mu\text{m}$  in diameter and 5 cm in length [45]. In the initial stage, the pinch plasma rapidly expanded and its radius increased by almost one order of magnitude. Sausage instability developing by 20 ns led to the axial stratification of the plasma column. Measurements of X-ray emission by using semiconductor diodes showed that the plasma temperature in the constrictions exceeded 10 keV. HXR emission detected in those experiments indicated the generation of runaway electrons in the plasma.

Thus, the results obtained in experiments with frozen deuterium fibers at currents close to the Braginskii–Pease current turn out to be very similar to those obtained in experiments with deuterated fibers. Both deuterated and frozen deuterium fibers do not provide good conditions for the efficient matching of the generator with the load (pinch), the formation of a uniform micrometer-diameter plasma column, and the concentration of the current to densities required to confine the plasma. Neutrons are mainly emitted during the development of sausage instability and the formation of hot spots. In experiments with deuterated and frozen deuterium fibers, the neutron yield was found to be anisotropic and to depend strongly of the current,  $Y \propto I^{5-10}$ . In the above papers, among mechanisms responsible for the generation of neutrons, thermal and acceleration mechanisms are considered. Their relative contributions depend on the experimental conditions: the fiber diameter, the current amplitude, the current rise rate, the energy input, and the presence of a prepulse.

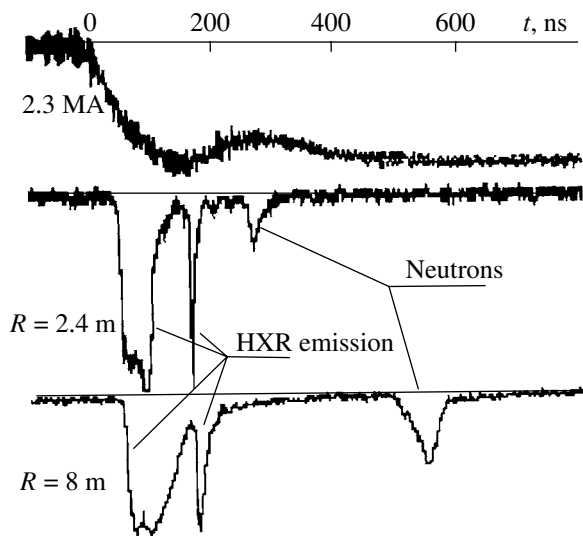
**2.2.3. Studies of neutron emission from profiled heterogeneous Z-pinch.** A new approach proposed in [41, 42, 57] to produce a hot plasma is based on the maximum amplification of a preformed perturbation. The small diameter of a hot plasma in a constriction arising due to the development of sausage instability is achieved at the expense of radial plasma compression and plasma outflow along the pinch axis [50, 51]. The above proposal is of interest, in particular, from the standpoint of igniting a thermonuclear burn wave that will propagate from the constriction of a dense Z-pinch over the entire region occupied by the pinch plasma. It was shown in [41, 42] that, in order to ignite a self-sustained thermonuclear reaction in a preformed con-



**Fig. 6.** Total yield  $Y$  of D–D neutrons from 125- $\mu\text{m}$  frozen deuterated fibers as a function of the discharge current  $I$ . The straight line corresponds to the dependence  $Y = 3.9 \times 10^{-6} I^5$  (where the current is in kA).

tion of a dense Z-pinch and to provide the subsequent expansion of the combustion region throughout the pinch, it is necessary that a current of 10 MA be passed through a 0.1-cm-diameter cylinder made of a condensed D–T mixture with a density of 0.2 g/cm<sup>3</sup>. Calculations performed in [41, 50] show that, in order that the combustion wave propagate along the pinch, it is necessary that the constriction diameter be a few micrometers, the plasma temperature in the constriction be higher than 2 keV, and the plasma density be about 500 g/cm<sup>3</sup>. For a higher current, the requirements for the final constriction diameter and plasma density are not so severe. For example, at a current of 20 MA achieved in the PBFA-Z facility ( $W = 11.4$  MJ,  $U = 2$  MV,  $\tau = 100$  ns) [58], the plasma density and the constriction radius needed for igniting a self-sustained thermonuclear reaction are equal to  $\rho = 50$  g/cm<sup>3</sup> and  $r_c = 20$   $\mu\text{m}$ , respectively [59].

In order to obtain a dense plasma in a pinch constriction, experiments with solid-state porous loads modeling deuterium loads were performed in [60, 61] at currents of a few megaamperes. With such loads, the initial density could be varied over a wide range, from the gas density ( $10^{-3}$  g/cm<sup>3</sup>) to the density of solids (0.7 g/cm<sup>3</sup>). The formation of the plasma in a dense Z-pinch was thoroughly investigated in the Module Angara 5-01



**Fig. 7.** Waveforms of the current and the intensities of neutron and HXR emission measured in the radial direction at distances of  $R = 2.4$  and  $8$  m from the pinch axis. The load is a 5-mm-diameter profiled deuterated agar-agar cylinder ( $\rho = 0.1$  g/cm<sup>3</sup>) with a neck 1 mm in diameter. The total neutron yield is  $Y = 7 \times 10^7$ .

facility ( $U = 100$  kV,  $I = 2.5$  MA,  $\tau = 100$  ns) [61–63] and in the S-300 eight-module generator ( $U = 700$  kV,  $I = 4$  MA,  $\tau = 70$  ns) [64–66]. The plasma in those experiments was produced by passing current pulses with an amplitude of up to 3 MA through profiled heterogeneous loads made of porous materials (agar-agar, polyvinylchloride, etc.). The mass density of the loads could be varied in the range 10–120 mg/cm<sup>3</sup>, the load diameter was 3–5 mm, and the length was 1 cm. In order to enhance the effect of energy focusing, a neck 0.1–1 mm in diameter was preformed in the central part of the load. In experiments performed in the Module Angara 5-01 facility, it was found that, when a current pulse with an amplitude of 1–1.5 MA was passed through a heterogeneous load with a mass density of 0.01 g/cm<sup>3</sup>, a dense hot plasma with a typical diameter of  $\sim 100$   $\mu$ m was produced in the region of a neck with an initial diameter  $< 1$  mm due to the development of a constriction. An analysis of X-ray line emission of highly charged ions showed that the electron temperature and plasma density in the constriction were  $T_e \sim 1$  keV and  $n_e = 10^{21}$  cm<sup>-3</sup>, respectively. The lifetime of the hot plasma was determined from the FWHM of the SXR pulse and was found to be  $\approx 5$  ns. Measurements of the current-channel diameter and the current density by using the Faraday rotation effect showed that all the current passed through an axial region with a radius of less than 1 mm.

The highest plasma parameters,  $n_e = 10^{22}$  cm<sup>-3</sup>,  $T_i \approx 6$  keV, and  $T_e = 1$ –1.5 keV, were obtained in the S-300 facility ( $I = 3$  MA,  $\tau = 100$  ns) by using profiled agar-

agar loads with a mass density of 0.1 g/cm<sup>3</sup> [64, 65]. To achieve neutron generation, a deuterium-containing substance (CD<sub>2</sub> or C<sub>14</sub>D<sub>10</sub>) was introduced in the load neck. The experiments showed that, at a current of 3 MA, the constriction radius decreased to 40–70  $\mu$ m, the compression ratio being  $R/r = 20$ , where  $R$  is the radius of the preformed neck and  $r$  is the final radii of the constriction. It was found that the ion temperature considerably exceeded the electron temperature. The ion temperature measured from the Doppler broadening of the emission lines of highly charged ions turned out to be  $T_i = 3$ –13 keV. Laser probing showed that the averaged velocity of the low-density peripheral plasma ( $n_e \leq 10^{19}$  cm<sup>-3</sup>) produced in the neck region in the current rise phase did not exceed  $2 \times 10^5$  cm/s.

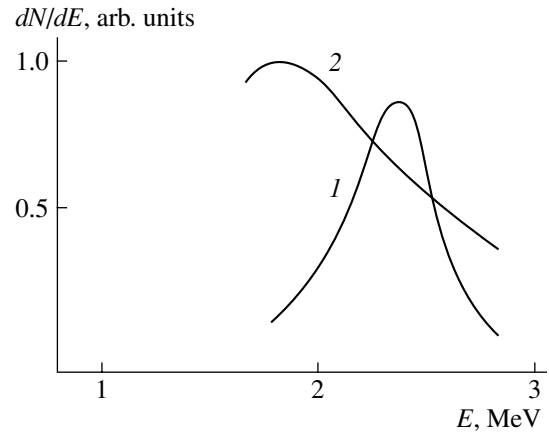
Fast-framing photography in the optical, VUV, and SXR spectral ranges revealed the effect of prolonged plasma production in the neck region. It took a few nanoseconds for short-term plasma fluxes to propagate from the neck surface to the load axis. Two-dimensional ideal-MHD simulations of a Z-pinch showed that the most important consequence of a prolonged plasma production in the neck region was that the current was intercepted by a newly produced plasma [65]. During the entire phase of plasma production, the current flowing through the area of radius 100  $\mu$ m did not exceed one-half of the total current. After plasma production terminated, almost all the current concentrated in this area for a few nanoseconds. This was accompanied by an abrupt increase in the plasma temperature. Prolonged plasma production stabilized the Z-pinch against the sausage instability. This effect provided the pinch to be stable while the pinch current increased to its maximum value.

The production of a hot plasma in the constriction of a dense Z-pinch was accompanied by the generation of SXR ( $E > 0.4$  keV) and HXR ( $E > 30$  KeV) emissions, which lasted for 3–4 ns and less than 10 ns, respectively [66]. A neutron pulse with a duration of 10 ns appeared simultaneously with the generation of X-ray emission (see Fig. 7). The neutron energy measured by the time-of-flight method both along the axis and at the right angle to it was  $2.5 \pm 0.3$  MeV [67], which corresponded to the D–D reaction. The maximum neutron yield from a single constriction was  $10^8$  neutrons per shot, which agreed with the theoretical estimate by the formula  $Y = (1/2)n^2\langle\sigma v\rangle_{dd}V\tau$  (where  $V$  is the volume of the hot plasma,  $\sigma$  is the cross section for the D–D reaction,  $v$  is the relative velocity of deuterons, and  $\langle\sigma v\rangle_{dd}$  is the reaction rate averaged over the velocity distribution) for a deuteron temperature of 6 keV, deuteron density of  $n_i = 10^{21}$  cm<sup>-3</sup>, typical hot-spot size of 100  $\mu$ m, and hot-spot lifetime of  $\tau = 4$  ns. The fact that short neutron, SXR, and HXR pulses appeared simultaneously indicates the formation of a single hot plasma spot in the neck region.

Thus, the experiments performed with profiled heterogeneous loads at currents of up to 3 MA and a current rise time of 100 ns indicate the development of a single constriction from a preformed neck. A hot plasma with parameters close to thermonuclear ones ( $n_e = 10^{22} \text{ cm}^{-3}$ ,  $T_i \approx 6 \text{ keV}$ , and  $r = 20 \text{ }\mu\text{m}$ ) was produced in the neck region. Fast-framing and time-integrated photography in VUV and SXR ranges show that the process of plasma production is prolonged. The formation of a hot dense plasma was accompanied by the generation of neutron and X-ray pulses with a duration of  $\leq 10 \text{ ns}$ . The fact that short neutron, SXR, and HXR pulses appear simultaneously confirms the hypothesis that they are generated from a hot plasma produced in a single constriction.

**2.2.4. Experiments with dense deuterium jets.** In contrast to conventional gas-discharge systems, which use capacitor banks with a relatively low charging voltage and allow one to generate millisecond current pulses at pressures of a few Torr and typical current-sheath radii of  $\leq 50 \text{ cm}$ , generators of high-power nanosecond current pulses make it possible to operate at considerably smaller initial radii ( $\sim 1 \text{ cm}$ ) and higher initial densities. This allows one to significantly increase the intensity of neutron emission generated during the formation of a superdense pinch under the action of a megaampere current pulse.

A high neutron yield (up to  $10^{12}$  neutrons per shot) was achieved in Europe's largest facility Angara-5-1 ( $U = 1.2 \text{ MV}$ ,  $I = 5 \text{ MA}$ ,  $\tau = 140 \text{ ns}$ ) in experiments on the compression of a dense deuterium jet injected into a 20- to 40-mm-long acceleration gap through a fast gas valve [68]. The compression of an axially nonuniform jet with an initial diameter of 15–30 mm and a linear number density of  $2 \times 10^{18} \text{ cm}^{-1}$  at the nozzle exit resulted in the formation of a pinch with a diameter of  $\sim 1 \text{ mm}$ , a length of 1–1.5 cm, and an electron temperature of  $T_e \approx 0.7\text{--}1 \text{ keV}$  in the acceleration gap. At the instant of maximum compression, the voltage across the diode dropped abruptly and a neutron pulse with a duration of 40–50 ns was generated. The average energy of neutrons emitted in the radial direction was 2.45 MeV, which corresponded to the D–D reaction. The average energy of neutrons emitted in the axial direction (from the cathode to the anode) was lower, 1.8–2.0 MeV. The FWHMs of the energy spectra of neutrons emitted in the radial and axial directions were 250–300 and  $\leq 600 \text{ keV}$ , respectively (see Fig. 8). An analysis of the anisotropic neutron energy spectrum showed that the neutrons were generated in the interaction with deuterons accelerated to an energy of 200–500 keV, the effective longitudinal and transverse temperatures being  $\sim 200$  and 20–50 keV, respectively. The low transverse temperature of the fast deuterons indicates that they were accelerated in the axial region, where the magnetic field was low. In [68], the acceleration of deuterons along the pinch axis was attributed to the anomalous resistance of the turbulent anode



**Fig. 8.** Energy spectrum of ions emitted in the (1) radial and (2) axial directions relative to the pinch axis.

plasma, across which a high voltage drop was generated.

An unsurpassed neutron yield,  $6 \times 10^{13}$  neutrons per shot, was achieved in the Z super-power facility ( $P_{\text{gen}} = 40 \text{ TW}$ ,  $U = 2 \text{ MV}$ ,  $I = 20 \text{ MA}$ ,  $\tau = 100 \text{ ns}$ ) in experiments on the compression of two hollow coaxial deuterium jets 1–2 and 2–4 cm in diameter [9]. The above value of the neutron yield agrees with the results of numerical simulations performed in that study. Such a high neutron yield allows one to suggest that, in the upgraded Z facility (the ZR facility), which will provide a current of 27 MA, a neutron yield of  $2 \times 10^{15}$  neutrons per shot will be achieved in experiments with similar loads (deuterium jets) and up to  $6 \times 10^{16}$  neutrons per shot in experiments with deuterium–tritium jets.

### 3. THEORY OF NEUTRON GENERATION FROM Z-PINCHES

After it was revealed that a pinch discharge in deuterium was accompanied by a neutron burst [1–6], a great deal of theoretical work has been done to describe this phenomenon [12–15].

First, neutron emission from Z-pinches was described by analogy with known mechanisms for generating neutrons in fusion reactions. At that time, two types of facilities were known in which intense fusion reactions occurred: deuterium tubes and thermonuclear bombs. In both of them, neutrons are generated due to the fusion of deuterium nuclei,  $D + D \rightarrow n + T$ , or deuterium and tritium nuclei,  $D + T \rightarrow n + \text{He}$ . A nuclear fusion reaction occurs when deuterium and tritium nuclei acquire an energy sufficient to overcome the electrostatic repulsion barrier, and the fusion energy is released in the form of fast particles, in particular, neutrons.

Mechanisms for generating neutrons in thermonuclear explosions and deuterium tubes differ only in the

processes by which the interacting particles acquire energy. In deuteron tubes, deuterium and tritium ions acquire energy by accelerating in the applied electric field, whereas in thermonuclear explosions, by colliding with hotter particles.

By analogy with thermonuclear explosions, the latter mechanism was called the thermonuclear mechanism, while the former came to be known as the acceleration mechanism.

It is well known that, in thermonuclear explosions, the ion energy distribution is close to Maxwellian, whereas in the acceleration mechanism, it is comprised of two groups of particles: low-energy (target) particles and fast particles, which were accelerated by the electric field. Accordingly, the acceleration mechanism is sometimes called the target mechanism. Neutron yields for these two types of distribution are somewhat different. Therefore, in searching new methods for increasing the neutron yield from Z-pinches, the role played by each of these mechanisms was investigated.

There are a lot of papers in which neutron emission from Z-pinches is explained by one of these mechanisms [14, 69–79]. Investigations have shown, however, that the ion energy distribution in Z-pinches, as well as the neutron emission spectrum from them, can neither be attributed to one of these mechanisms nor be described as a result of their combined action [13].

When analyzing neutron emission from Z-pinches, we will consider three main problems disputed over a long time in the scientific literature: how ions in the pinch plasma acquire energy, what is their energy spectrum, and why neutron emission is anisotropic.

### 3.1. Mechanisms for Generating Fast Ions in Z-Pinches

**3.1.1. Acceleration of ions in the longitudinal electric field of a pinch.** For the generation of fusion neutrons, it is necessary that there be a sufficient number of fast ions. The first hypothesis on the origin of fast ions in Z-pinches was that they were accelerated in the axial electric field [71–76]. This hypothesis was based on the fact that, at the instant corresponding to a singularity in the current waveform, a high ohmic electric field arises, in which charged particles can acquire a considerable energy.

However, the hypothesis of ion acceleration in the ohmic electric field turned out to be wrong, because the ions have no time to acquire a sufficient energy between two successive collisions. In other words, they are accelerated too slowly and lose their energy in collisions too quickly.

The electrons acquire energy in the longitudinal electric field much more rapidly than do the ions. However, due to Coulomb collisions, they also lose almost all the momentum they acquire in the electric field. Only a small fraction of electrons is accelerated to high energies due to the runaway mechanism [80], which

consists in that, if an electron has acquired a high velocity, then the cross section for its interaction with other charged particles decreases substantially, so it can be freely accelerated by the electric field.

There is no analogous mechanism for ions, because the characteristic ion velocity in a discharge is substantially lower than the electron velocity. Therefore, the effective electron–ion collision frequency is only determined by the electron velocity and does not depend on the ion velocity. As a result, the typical ion deceleration time turns out to be the same for all ions. The fact that the ion frictional force is independent of their velocity does not allow any group of ions to acquire energy higher than the energy of the bulk ions.

**3.1.2. Ohmic heating.** In the early fusion studies (including experiments with high-power pulsed discharges), ohmic heating was supposed to be the main mechanism for heating the discharge plasma [81]. Along with neutral injection, ohmic heating is the main mechanism for plasma heating in tokamaks. A characteristic feature of ohmic heating is that the electromagnetic field energy is first deposited in the electron component and then is transferred to ions via electron–ion collisions.

It turned out, however, that ohmic heating plays a significant role only in quasi-steady systems. In pinches, this mechanism is of minor importance because of the short heating time and high plasma conductivity. In Z-pinches, it becomes important only in the decay phase, in which the plasma resistance is anomalously high and the neutron emission intensity usually decreases.

**3.1.3. Adiabatic compression of the Z-pinch plasma column and formation of hot plasma spots in constrictions.** Experimental studies show that a considerable fraction of neutron emission from the pinch plasma is related to the development of MHD instabilities, in particular, sausage instability. If the plasma radius in a certain cross section becomes less than that in the neighboring regions, then the magnetic field in this cross section will increase. The increased magnetic field further squeezes the plasma in the constriction region, and the plasma temperature there grows. The high plasma pressure in the constriction leads to the axial plasma outflow from this region (Fig. 9). This results in a further increase in the plasma temperature in the constriction.

The development of a constriction can easily be described in a zero-dimension model [82–84]. Let  $N$  be the linear plasma density in the constriction and  $\tau$  be the time during which the plasma flows out of the constriction. Then, the plasma escape from the constriction is described by the equation

$$\frac{dN}{dt} = -\frac{N}{\tau}. \quad (3.1)$$

Experiments show that the time  $\tau$  is proportional to the ion transit time along the pinch radius,

$$\tau = \frac{\alpha r}{v_s}, \quad (3.2)$$

where  $r$  is the constriction radius,  $\alpha$  is a numerical factor (typically,  $\alpha \sim 10$ ), and  $v_s$  is the ion acoustic velocity in the constriction region. The presence of a relatively large numerical factor is related to the fact the plasma first flows along the axis up to the constriction boundary and then escapes along the radius [82].

A decrease in the number of particles in the constriction leads to its squeezing and to an increase in the plasma density and temperature within it. The plasma temperature in the constriction can reach  $\sim 10$  keV, and the plasma density is much higher than the initial one. In the final stage of plasma compression, the plasma pressure in the constriction is determined by the pressure of the azimuthal magnetic field; therefore, Bennet condition (1.1) for radial equilibrium is satisfied with a good accuracy. As the plasma flows out of the constriction region, the linear plasma density  $N$  in the constriction decreases and the plasma temperature increases according to formula (1.1). When the temperature becomes sufficient for initiating nuclear fusion reactions, the constriction region becomes a source of intense neutron emission.

The dynamics of the plasma and neutron emission from a constriction can be described using the equation of plasma energy balance in the constriction,

$$2\frac{2dNT}{3} = -\frac{52NT}{2\tau} - 2\pi r \frac{B^2}{8\pi} \frac{dr}{dt}. \quad (3.3)$$

The first term on the right-hand side describes the convective escape of the plasma energy from the constriction region, while the second term describes the work done by the magnetic field to compress the plasma. If the current  $I$  is constant (i.e.,  $NT \sim I^2 = \text{const}$ ), then Eqs. (3.1)–(3.3), together with Bennet condition (1.1), have a simple analytic solution describing the time dependence of the plasma parameters in the constriction:

$$\begin{aligned} \tau &= \tau_0(1 - t/t_1), \\ N &= N_0(1 - t/t_1)^{4/7}, \\ T &= T_0(1 - t/t_1)^{-4/7}, \\ r &= r_0(1 - t/t_1)^{5/7}, \\ n &= n_0(1 - t/t_1)^{-6/7}. \end{aligned} \quad (3.4)$$

It follows from this solution that the plasma temperature  $T$  and density  $n$  in a constriction increase without bound as  $t \rightarrow t_1$ . The total neutron yield from the pinch can be obtained by integrating the neutron intensity  $n^2 \langle \sigma v \rangle \pi r^2 h$  over the time. Although the tempera-

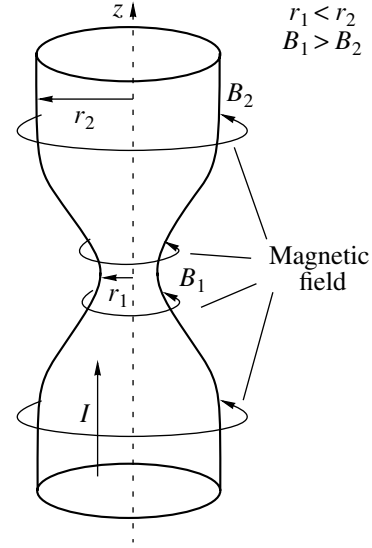


Fig. 9. Development of a Z-pinch constriction and formation of a hot plasma in the constriction.

ture and density increase without bound as  $t \rightarrow t_1$ , the neutron yield is finite. For the D–D reaction, it is equal to [82, 83]:

$$Y_{dd} = 3.62 h \alpha n_0^{1/2} T_0^{-3/4} I^3, \quad (3.5)$$

where  $I$  is the pinch current (in A),  $n_0$  is the plasma density (in  $\text{cm}^{-3}$ ), and  $T_0$  is the plasma temperature (in eV) at a certain initial instant (e.g., the instant at which the shock wave arrives at the pinch axis).

A comparison with experiments shows that expression (3.5) adequately describes not only the absolute value of the neutron yield but also its dependence on the discharge parameters [85, 86]. This means that a considerable part of neutron emission can be ascribed to the adiabatic compression of the pinch plasma and to the formation of hot plasma regions in the constrictions.

**3.1.4. Ion acceleration due to the axial plasma outflow from a constriction.** The formation of a hot plasma leads to the emergence of axial plasma jets on the both sides from the region of maximum compression. These jets are sources of fast ions. If we suppose that the pinch plasma parameters correspond to solution (3.4), the plasma particles in the constriction obey a Maxwellian distribution, and the plasma outflow from the constrictions is described by Eq. (3.1), then the time-integrated spectrum of ions  $n_i$  escaping from the constriction will have the form [85]

$$\frac{dn_i}{d\varepsilon} = \frac{3}{8c^2} \frac{h I^2}{\varepsilon^2}, \quad (3.6)$$

where  $h$  is the constriction height and  $\varepsilon$  is the energy with which ions leave the constriction. When the anomalous conductivity of the pinch plasma is taken into

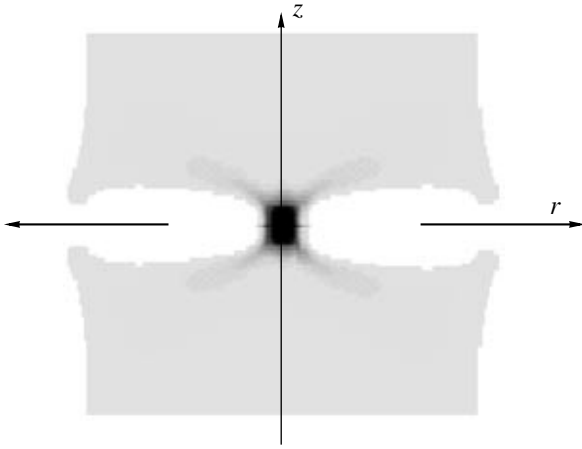


Fig. 10. Formation of a fast ion jet in a Z-pinch constriction.

account, the compression is weaker and the spectrum of ions escaping from the pinch is somewhat different. In this case, the ion spectrum depends on the energy as  $\varepsilon^{-k}$ , where  $k$  is in the range 2–3 [85].

In [76], a self-similar solution describing the development of a constriction in a so-called “hydraulic” model was derived. It was assumed that the density, temperature, and axial velocity of the plasma inside the constriction depend only on  $z$  and are independent of the radius and that the radial velocity depends linearly on the radius. The self-similar solution was used to calculate the spectrum of ions escaping with a hydrodynamic velocity from the pinch constriction. In this model, the spectrum of the accelerated ions as function of the energy  $\varepsilon$  is described by the formula

$$\frac{dn_i}{d\varepsilon} \propto \varepsilon^{-(1+\sqrt{3})} \approx \varepsilon^{-2.7}. \quad (3.7)$$

Thus, in the hydraulic model, the spectrum of the accelerated ions also decreases with energy according to a power law, the exponent being again in the range 2–3.

The energy deposited in the fast particles is determined by the thermal energy of the pinch plasma; i.e., it is equal to  $W = 750hI^2$ . Here, the height of the pinch constriction is in centimeters, the pinch current  $I$  is in megaamperes, and the deposited energy  $W$  is in joules. For Z-pinch facilities, this energy is usually a few percent of the capacitor-bank energy.

The ions escaping from the pinch constriction collide with those in the rest part of the pinch, giving rise to neutron emission. This emission is additional to that emitted from the constriction region. Unfortunately, the integral neutron yield due to this mechanism is still poorly studied. Since the neutron yield depends strongly on the processes of ion deceleration in both the constriction and the surrounding plasma, the problem of determining the additional neutron yield is rather

complicated and requires a detailed calculation of ion trajectories. Nevertheless, estimates show that this yield can be comparable with the total neutron yield observed in PF facilities at moderate (<1 MA) discharge currents.

**3.1.5. Generation of fast ions in the plasma surrounding the pinch column.** In the simulations performed in [86, 87], the development of a pinch constriction was accompanied by the generation of fast ions with energies higher than 1 MeV. The source of these ions was the low-density plasma surrounding the dense pinch core.

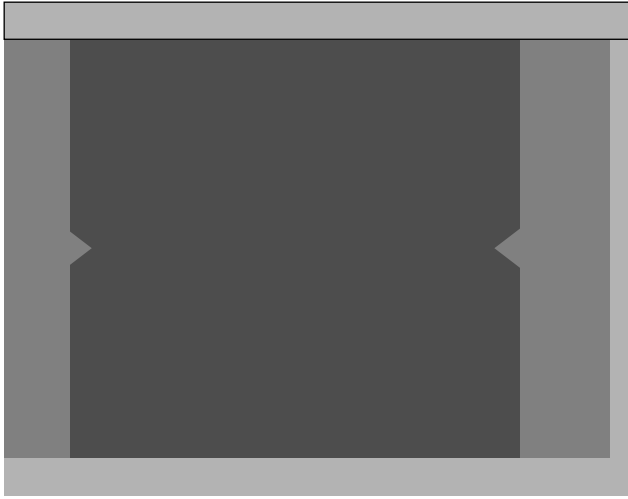
This effect can easily be explained by considering the development of a vacuum cavity in the pinch plasma as an example. The matter is that the pinch boundary is unstable against short-scale perturbations. The development of short-scale perturbations leads to the formation of a rapidly expanding cavity in the pinch column. This cavity is filled with a low-density plasma and the magnetic field frozen in it. The cavity expands in all directions, the typical expansion velocity being on the order of the Alfvén plasma velocity,  $V_A = B/\sqrt{4\pi\rho}$ . The cavity volume increases abruptly, but the plasma (and the magnetic field frozen in it) can enter the cavity only through a narrow slit (Fig. 10). Therefore, the velocity with which the magnetic field flows into the cavity can reach a very high value.

Let  $\Delta$  be the slit width and  $R_{\text{cav}}$  be the mean radius of the cavity. Then, in order for the cavity to be steadily filled with a magnetic field, the velocity with which the plasma flows into the cavity should be  $V_{\text{in}} = V_A \frac{2\pi R_{\text{cav}} - \Delta}{\Delta}$ . As a rule,  $\Delta \ll R_{\text{cav}}$ ; therefore, this velocity exceeds the Alfvén velocity by a factor of  $2\pi R_{\text{cav}}/\Delta$ .

Sometimes, because of the plasma inertia, the cavity is filled with a magnetic field too slowly. Then, at the entrance to the cavity, a current is induced, which shunts the main pinch current. This current flows across the azimuthal magnetic field, thereby pushing the low-density plasma into the cavity.

The velocity with which the magnetic field flows into the cavity is limited from above only by the inertia of the plasma frozen in this magnetic field. Thus, the ions entrained by the magnetic field enter this cavity with a velocity on the order of the Alfvén velocity for the low-density peripheral plasma. This velocity is fairly high and, for a pinch current of 1 MA, amounts to  $\sim 10^9$  cm/s, which corresponds to a deuteron energy of  $\sim 1$  MeV.

The energy of these ions considerably exceeds the energy of ions in the axial jets emerging from the pinch constrictions. However, these ions are fewer in number than those in the jets and their contribution to the neutron yield is usually small.



**Fig. 11.** Initial density distribution in simulations of the development of a pinch constriction.

### 3.2. Spectra of Fast Ions Obtained in MHD Simulations of a Z-Pinch

The main reason why the fast ions in a pinch have a power-law distribution over energies is the turbulent mechanism for their acceleration. Particular manifestations of such a turbulent mechanism are the formation of jets (see Section 3.1.4) and the radial acceleration of ions (see Section 3.1.5).

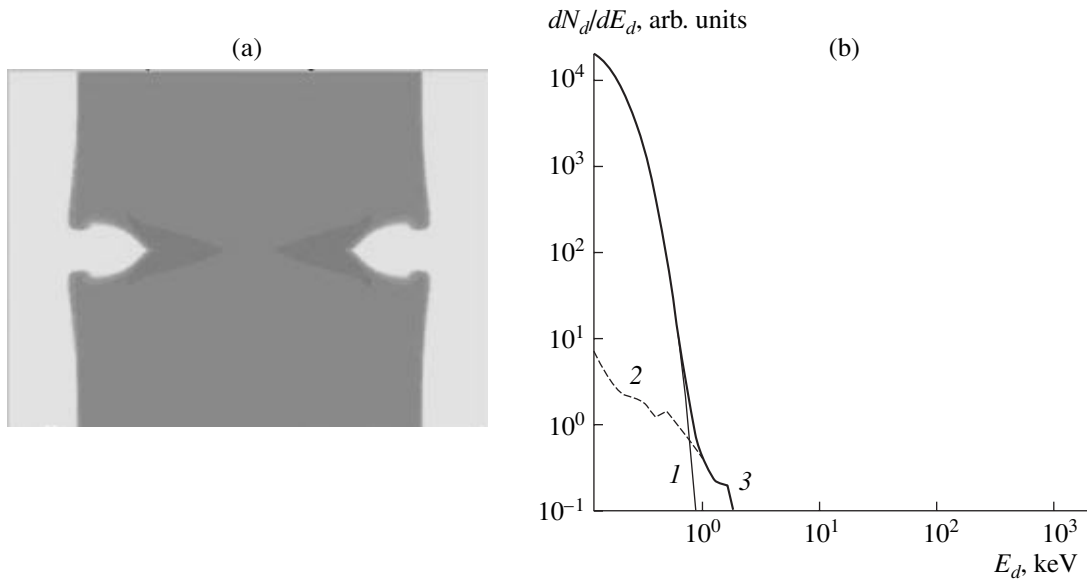
The development of the pinch turbulence can be analyzed using two-dimensional MHD simulations of the pinch dynamics. The ion velocity distribution in the

pinch plasma can be deduced from the data on the plasma temperature and plasma velocity. As a result, one can determine the time evolution of the ion energy distribution in the pinch plasma. Note that the energy spectrum thus obtained is not a spectrum of ions in a particular region of the pinch plasma but is the total energy spectrum of ions escaping from the pinch.

In [87, 88], an equilibrium plasma column whose gas-dynamic pressure was balanced by the magnetic pressure of the pinch current was chosen as an initial condition for simulations of two-dimensional pinch turbulence. It was assumed that, at the instant  $t = 0$ , there was a boundary perturbation in the middle of this pinch (Fig. 11). In this case, sausage instability develops and a cavity forms in the nonlinear stage of instability (Figs. 12–14). Further, the entire plasma column becomes turbulent and the magnetic field is intermixed with the plasma.

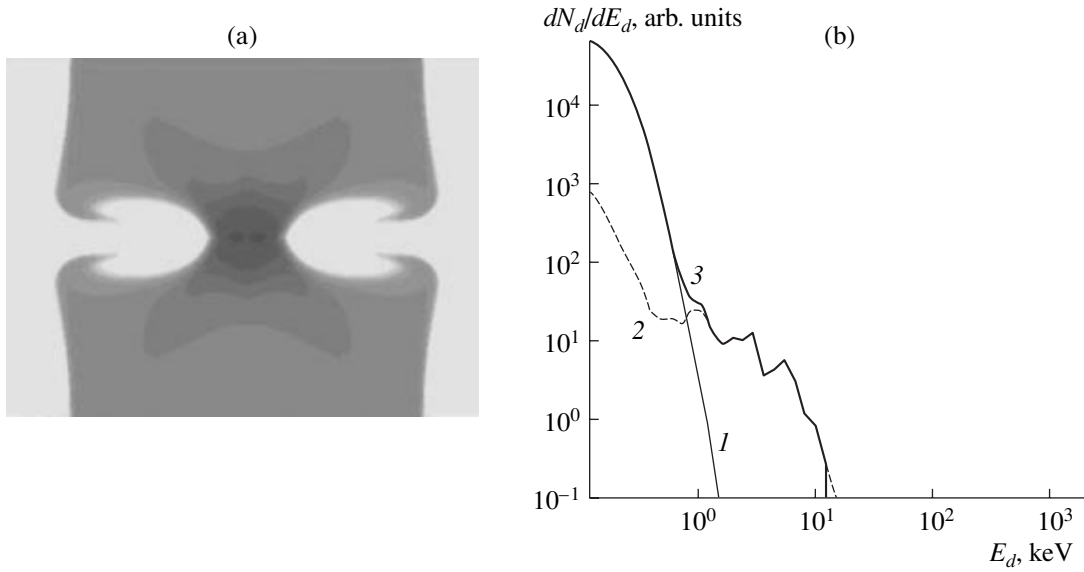
Using the simulation data, the velocity spectrum of the pinch particles and the energy spectrum of ions escaping from the pinch were determined. In order to find the integral ion energy spectrum in the pinch, the ions were assumed to have a Maxwellian velocity distribution in the reference frame moving with the plasma. Then, in the laboratory frame, the distribution of the plasma ions at each point is described by a Maxwellian distribution (with a temperature  $T(t, r, z)$ ) shifted in velocity space by the hydrodynamic plasma velocities  $v_z(t, r, z)$  and  $v_r(t, r, z)$ .

Let  $f(t, \epsilon, r, z)$  be the ion energy distribution at a point with coordinates  $r$  and  $z$ . Integrating this distribu-

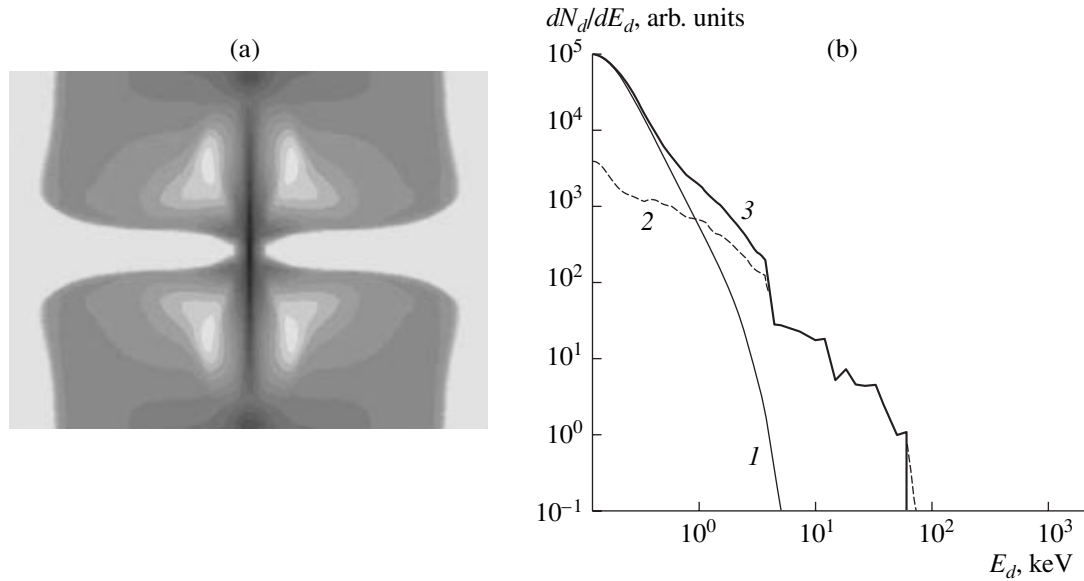


**Fig. 12.** (a) Distribution of the plasma density in a Z-pinch in the initial phase of MHD instability and (b) ion energy spectra calculated for the same instant with allowance for (1) only thermal ion motion, (2) only hydrodynamic motion, and (3) both thermal and hydrodynamic motion.





**Fig. 13.** (a) Distribution of the plasma density in the stage of the cavity formation in a pinch plasma column and (b) ion energy spectra calculated for the same instant with allowance for (1) only thermal ion motion, (2) only hydrodynamic motion, and (3) both thermal and hydrodynamic motion.



**Fig. 14.** (a) Distribution of the plasma density at the instant corresponding to the formation of an axial jet and (b) ion energy spectra calculated for the same instant with allowance for (1) only thermal ion motion, (2) only hydrodynamic motion, and (3) both thermal and hydrodynamic motion.

tion over the pinch volume, we obtain the averaged energy distribution of the pinch ions at a given instant  $t$ ,

$$F(t, \varepsilon) = \iint_V f(t, \varepsilon, r, z) 2\pi r dr dz. \quad (3.8)$$

In pinch experiments, the time-integrated ion energy spectrum is usually measured. Therefore, in order to

compare the simulation results with experiment, calculated ion spectrum (3.8) was integrated over time:

$$F_s(t, \varepsilon) = \int_0^t F(t', \varepsilon) dt'. \quad (3.9)$$

Curves 1 in Figs. 12–15 show the ion energy distributions  $F_s(t, \varepsilon)$  obtained in MHD simulations with

allowance for the thermal motion alone. In this case, the influence of the hydrodynamic plasma velocity on  $F_s(t, \varepsilon)$  was ignored; i.e., the integration was performed for the zero the azimuthal and radial plasma velocities ( $v_r(t, r, z) = 0$  and  $v_z(t, r, z) = 0$ ).

Curves 2 show the ion energy distributions obtained by integrating expression (3.8) with allowance for only the hydrodynamic plasma motion (the thermal ion motion was ignored).

Finally, curves 3 show the ion energy distributions  $F_s(t, \varepsilon)$  obtained with allowance for both the thermal ion motion and the hydrodynamic plasma velocities.

Figure 13 shows the spectrum of deuterons calculated for PF discharges with a current of  $\sim 1$  MA. In the energy range 3–1000 keV, the ion spectrum decreases with energy as  $f \sim \varepsilon^{-2}$ . The fast ions are produced both inside the plasma column and in the surrounding low-density plasma. In the range of higher energies, the number of particles decreases as  $f \sim \varepsilon^{-4}$ .

Thus, MHD simulations allow one to explain the spectra of ions emitted from Z-pinch. An analysis of the simulation results shows that the main mechanism by which the ions acquire energy is plasma acceleration under the action of the  $\mathbf{j} \times \mathbf{H}$  force. It is this force that accelerates the plasma and ions in all the pinch systems. The plasma is accelerated very differently in different plasma regions. This is why the accelerated ions were observed to have smooth energy spectra.

### 3.3. Dependence of the Neutron Yield on the Deuteron Energy Spectrum

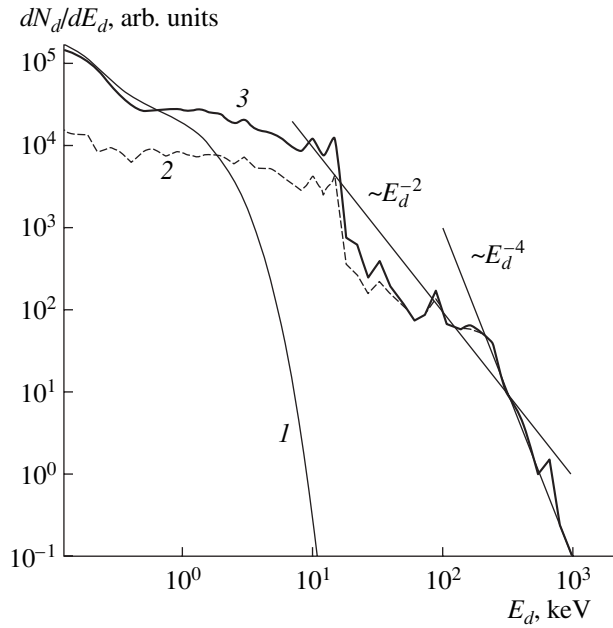
The separation of the mechanisms for neutron emission into the target and thermal ones provides two different ways of calculating the rates of nuclear fusion reactions. The neutron yield can easily be determined when either hot plasma ions have a Maxwellian energy distribution or the plasma is cold, but there is a fast monoenergetic ion beam.

In actual Z-pinch plasmas, neither of these two limiting situations occur. In the range of high energies, the ion distribution deviates from a Maxwellian one. On the other hand, no monoenergetic ion beams were observed in the pinch plasma.

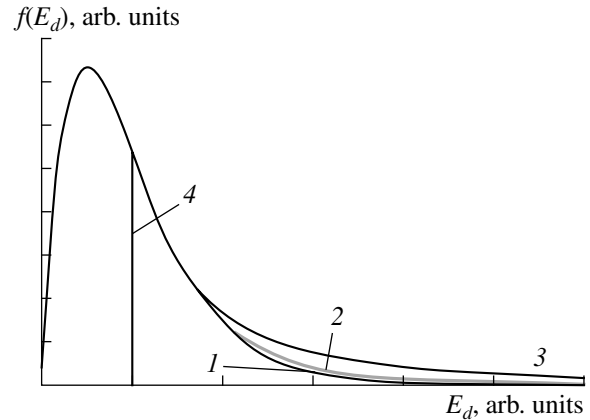
The intensity of neutron emission was analyzed in [89] for ion distributions that are Maxwellian at low energies ( $\varepsilon_d < (k+1)T_{\text{eff}}$ ) and have a power-law tail in the energy range  $\varepsilon_d > (k+1)T_{\text{eff}}$  (Fig. 16).

The rate of a thermonuclear reaction is determined by its cross section  $\sigma(\mathbf{V}_1 - \mathbf{V}_2)$  and the ion velocity distribution of  $f(\mathbf{V})$  [83],

$$\langle \sigma v \rangle = \iint f(\mathbf{V}_1) f(\mathbf{V}_2) |\mathbf{V}_1 - \mathbf{V}_2| \times \sigma(\mathbf{V}_1 - \mathbf{V}_2) d^3 \mathbf{V}_1 d^3 \mathbf{V}_2. \quad (3.10)$$



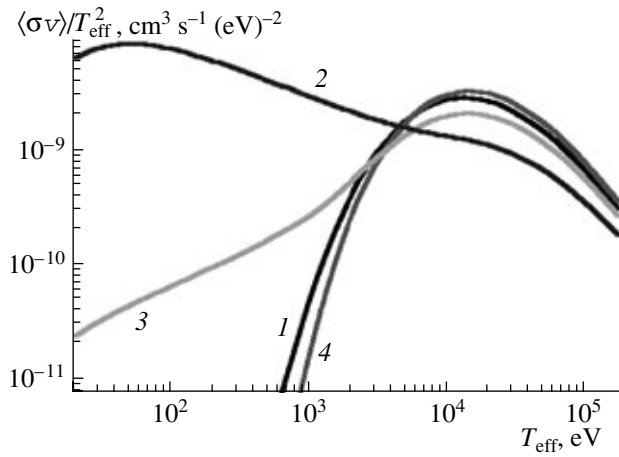
**Fig. 15.** Ion energy spectra in the phase of strong pinch turbulence, calculated with allowance for (1) only thermal ion motion, (2) only hydrodynamic motion, and (3) both thermal and hydrodynamic motion.



**Fig. 16.** Ion distribution functions with different tails: (1) a Maxwellian tail; (2) a power-law tail,  $f \sim 1/E_d^2$ ; (3) a power-law tail,  $f \sim 1/E_d^3$ ; and (4) a truncated (tailless) distribution function.

Figure 17 shows the rate of the D–D reaction at a constant pressure ( $nT_{\text{eff}} = \text{const}$ ) for different power-law tails of the ion distribution function. For comparison, the figure also shows the reaction rate for a Maxwellian ion distribution function.

For an effective ion temperature of  $T_{\text{eff}} > 3$  keV, the rate of a thermonuclear reaction in a plasma with a purely Maxwellian ion distribution differs slightly from



**Fig. 17.** Nuclear reaction rate as a function of the effective temperature at a constant plasma pressure for ion distribution functions with different tails: (1) a Maxwellian tail; (2) a power-law tail,  $f \sim 1/E_d^2$ ; (3) a power-law tail,  $f \sim 1/E_d^3$ ; and (4) a truncated (tailless) distribution function.

that in a plasma in which the ion energy distribution has a power-law tail.

For a temperature of  $T_{\text{eff}} < 3$  keV, the reaction rate is higher for ion distributions with a power-law tail. At temperatures in the range  $T_{\text{eff}} < 1$  keV, the nuclear reaction rate is expressed as

$$R \propto (nT_{\text{eff}})^2 T_{\text{eff}}^{k-2.5}. \quad (3.11)$$

The lifetime of the pinch plasma is determined by the ion transit time along the pinch radius,  $\tau = r/v_s$ , where  $v_s$  is the ion-acoustic velocity and  $r$  is the radius of the pinch plasma column. Therefore, the duration of the neutron pulse increases with decreasing effective temperature. As a result, the total neutron yield depends on the temperature as  $Y = RV \sim T_{\text{eff}}^{k-3}$ , where  $V$  is the plasma volume. This means that, for  $k = 3$ , the neutron yield is independent of the effective temperature, whereas for  $k = 2$ , it is inversely proportional to the temperature.

### 3.4. Neutron-Yield Scaling in Z-Pinches

The total neutron yield is obtained by integrating neutron emission intensity (3.10) over time and over the volume occupied by the pinch plasma. For a cylindrically symmetric pinch column, the neutron yield is expressed as

$$Y = \iiint n^2(r, z, t) \langle \sigma v \rangle(r, z, t) 2\pi r dr dz dt, \quad (3.12)$$

where  $r$  and  $z$  are the radial and axial coordinates and  $t$  is the time. It follows from Eqs. (3.10) and (3.12) that the neutron yield is determined by the ion energy distribution,

the particle density squared, the volume occupied by plasma, and the plasma lifetime.

The simplest energy scaling for the neutron yield can be obtained by assuming that the ion energy distribution does not depend on the input energy. In this case, the plasma lifetime is determined only by the plasma dimensions (is proportional to the plasma radius).

Let us first consider the case where the configuration and dimensions of the plasma are independent of the input energy and only the plasma density varies. Then, the lifetime of each element of the plasma volume is also independent of the input energy. It follows from (3.12) that, in this case, the neutron yield is proportional to the plasma density squared ( $Y \propto n^2$ ). Since, in Z-pinches, the energy per unit of the pinch height is proportional to the current squared ( $W \sim I^2$ ) and all this energy is spent on increasing the plasma density ( $n \sim W$ ), the neutron yield depends on the current in the fourth power,

$$Y \sim n^2 \sim W^2 \sim I^4. \quad (3.13)$$

This scaling is most often used to estimate the neutron yield in the case of thermal neutron emission, when plasma is in thermodynamic equilibrium. Nevertheless, this scaling is also valid for any other ion energy distributions, provided that the shape of the distribution does not depend on the energy deposited in the plasma.

Scaling (3.13) assumes that the initial mass density of the compressed pinch substance increases with increasing current. However, in gaseous Z-pinches and PFs, the pressure and the initial gas density cannot be increased substantially without changing the formation conditions of the plasma sheath. Therefore, for such pinches, the neutron scaling is somewhat different.

If we assume that an increase in the input energy does not affect the plasma density and the energy distribution over the plasma column and leads only to a self-similar increase in the plasma dimensions ( $z_0 \sim r_0 \sim \tau_0 \sim W^{1/3}$ ), then the neutron scaling is expressed as

$$Y \sim z_0 r_0^2 \tau_0 \sim W^{4/3} \sim I^{8/3} \sim I^{2.7}. \quad (3.14)$$

For gaseous pinches, the situation is somewhat different: the pinch dimensions for the same discharge chamber can be increased only by increasing in pinch radius, while the height of the compression region,  $z_0$ , is kept unchanged. In this case, for the pinch radius, we have  $r \sim t \sim W^{1/2}$  and the neutron yield is

$$Y \sim z_0 r_0^2 \tau_0 \sim W^{3/2} \sim I^3. \quad (3.15)$$

If an increase in the pinch current leads to an increase in both the plasma density and the geometrical dimensions of compressed pinch regions, then the neutron yield increases with increasing current as  $I^k$ , where the exponent  $k$  is in the range 3–4. Such a dependence is characteristic of PF facilities.

In principle, the exponent in the current scaling for the neutron yield can be larger than 4. This is the case if the ion energy spectrum varies with increasing input energy in such a way that a fraction of ions having the optimal energy for the nuclear reaction (100–400 keV) increases. However, an abrupt increase in the neutron yield due to a change in the ion energy spectrum can take place only in a narrow energy range. Such scalings are usually observed in low-energy systems.

### 3.5. Anisotropy of Neutron Emission

As was mentioned in Section 2, there are many papers in which much attention was paid to analyzing mechanisms that give rise to the neutron emission anisotropy [19, 90–92]. The anisotropy of neutron emission from Z-pinches implies that the distribution function of the colliding ions is anisotropic. One mechanism for the appearance of such an anisotropy is that the ions accelerated in the axial electric field of the pinch interact with slow ions of the low-temperature pinch plasma. However, the measured anisotropy of neutron emission turned out to be much less than it would be if the neutron emission were caused only by the axial ion beam.

In the 1970s, the model of a moving nuclear pile [19] was proposed to explain the anisotropy of neutron emission from pinches. Imshennik and D'yachenko [92] analyzed mechanisms for neutron generation by using two-dimensional hydrodynamic simulations. They found that the velocity of the emerging plasma jets was very high and estimated the neutron yield from these jets. This yield turned out to be comparable with the total neutron yield from a discharge. From the standpoint of neutron generation, the axial plasma jet acts as a moving nuclear pile. In this case, the anisotropy of neutron emission from a Z-pinch can be explained by the motion of the thermonuclear source (i.e., the jet) along the pinch axis.

The further analysis showed, however, that the jets were ejected from the constrictions in both direction, toward the cathode and the anode. Therefore, the neutron anisotropy that takes place in one jet should be compensated for by the anisotropy of neutrons emission from the jet ejected in the opposite direction. Hence, the presence of two counterpropagating jets suppresses or substantively decreases the anisotropy of neutron emission.

The most detailed theoretical analysis of the neutron emission anisotropy in pinch discharges was given in [93]. It was shown that the anisotropy of ion distribution is an inherent feature of Z-pinches. Even if the ion distribution in a pinch is isotropic at a certain instant, the more energetic ions accelerated by any mechanism will deviate farther from the pinch axis, moving in the transverse magnetic field. This field deflects fast ions to the cathode, so they begin to move preferentially toward the cathode. This effect is quite sufficient to

explain the observed anisotropy of the colliding ions and the resulting anisotropy of neutron emission from Z-pinches. Simple estimates show that Coulomb collisions have no time to make the ion distribution isotropic.

## 4. IGNITION OF A THERMONUCLEAR REACTION IN Z-PINCHES

At present, the generation of neutron emission in Z-pinches is interpreted as follows. When a high current flows through a pinch, the plasma column is compressed under action of the magnetic field and hot dense plasma bunches are produced at the pinch axis. These plasma bunches, which are most pronounced in the pinch constrictions, are sources of thermal neutron radiation.

Moreover, the turbulent pinch plasma and the hot plasma produced in the pinch constrictions are sources of fast ions. Interacting with deuterium ions of the adjacent plasma regions, these ions generate additional neutron emission.

In investigations aimed at achieving controlled thermonuclear fusion, the Lawson criterion [94] is usually implied as a necessary condition for thermonuclear ignition.

However, the condition for thermonuclear ignition differs from the Lawson criterion, which is in fact a condition for a thermonuclear fusion reaction to be self-sustained. The nuclear reaction is sustained due to energy transfer from the charged reaction products to deuterium and tritium ions, which then enter into fusion reactions. In order for this energy transfer to be efficient, it is necessary that the collisions between plasma particles be sufficiently frequent. The high collision frequency between the particles involved in thermonuclear burn makes the ion distribution close to a thermal one.

On the other hand, in the stage of thermonuclear ignition, the ion distribution may be nonthermal. The ions of the igniting plasma usually acquire the energy directly from the electromagnetic field rather than from other particles. Therefore, their distribution can differ substantially from a thermal one. The Lawson criterion was written in terms of the temperature and the confinement parameter  $n\tau$ . In contrast, the ignition criterion cannot be expressed in these terms because the notion of the temperature can be inapplicable to the igniting plasma.

Z-pinches were the first plasma systems in which neutron emission was observed, whereas the ions had a nonthermal distribution. The ions in the pinch plasma acquire energy due to the acceleration in the electric field rather than in collisions with other particles. Collisions have no time to thermalize the Z-pinch plasma completely, so the distribution of the plasma ions differs from a thermal one. Such a non-Maxwellian plasma turned out to be more appropriate for fusion

reactions than a plasma with a thermal ion distribution [95].

Obviously, if one produces a plasma in which the Lawson criterion for a fusion reaction to be self-sustained is satisfied, then this plasma will ignite a thermonuclear chain reaction in the system. However, thermonuclear ignition can be achieved without producing a thermal plasma. This substantially facilitates the ignition problem. A nonthermal plasma is easier to ignite because fast ions can be produced in the igniting plasma not only due to collisional heating but also by accelerating the deuterium and tritium ions to high energies in electric fields. In this case, their distribution should not be necessarily thermal. Moreover, it is possible to produce such an ion energy distribution at which fusion reactions will proceed more intensely than in the case of a thermal ion distribution.

The Lawson criterion, which is based on the notion of the temperature, is inapplicable to nonthermal plasmas. Nevertheless, there is a criterion for thermonuclear ignition in such plasmas. This criterion is related to the condition for achieving a necessary yield and intensity of neutron emission from the system [95, 96].

The first requirement for a thermonuclear burn wave to be generated in a deuterium–tritium Z-pinch is to heat some plasma region so that the energy released in the charged reaction products (alpha-particles) be equal to or higher than the thermal energy of the plasma that is the source of these reaction products, i.e., so that the following condition be satisfied:

$$Q_\alpha > Q_{\text{plasma}} \quad (4.1)$$

The thermal plasma energy in the pinch constriction,  $Q_{\text{plasma}}$ , can be estimated from the Bennet condition,  $\pi r^2 n T = I^2 / 4c^2$ . It follows from the latter condition that the thermal energy in the Z-pinch plasma column of length  $h$  is equal to  $Q_{\text{plasma}} = 750 I^2 h$ , where  $Q_{\text{plasma}}$  is joules,  $h$  in centimeters, and the current  $I$  is in megaamperes. Taking into account that, in the D–T reaction, the energy released in alpha-particles is equal to  $\epsilon^* = 3.5$  MeV per neutron, inequality (4.1) can be represented in the form

$$Y_{\text{DT}} > 750 h I^2 / (1.6 \times 10^{-19} - 3.5 \times 10^6) = 1.34 \times 10^{15} h I^2, \quad (4.2)$$

where  $Y_{\text{DT}}$  is the neutron yield of the D–T reaction ( $Y_{\text{DT}} = Q_\alpha / \epsilon^*$ ),  $h$  is the length of the region emitting neutrons (in cm),  $I$  is the current (in MA), and  $1.6 \times 10^{-19}$  is the electronvolt-to-joule conversion constant. Expression (4.2) determines the neutron yield at which a thermonuclear burn wave in a D–T mixture starts to propagate along the Z-pinch.

Since the thermonuclear burn wave propagates with a velocity of  $10^8$  cm/s, the intensity of neutron emission should be

$$dY_{\text{DT}}/dt = Y_{\text{DT}} \times 10^8 / h > 1.3 \times 10^{23} I^2, \quad (4.3)$$

where the current  $I$  is in megaamperes.

Conditions (4.2) and (4.3) were derived for a thermonuclear burn wave propagating along a Z-pinch; therefore, they should also be satisfied for the plasma igniting this wave.

Besides these two necessary conditions for igniting the D–T reaction, the third condition should also be satisfied; namely, the charged reaction products should transfer their energy to the plasma within a sufficiently small region near the place where they were produced. In this case, the energy of alpha particles will be completely utilized to initiate a thermonuclear burn wave.

The third condition implies, first, that the charged products of the fusion reaction should be magnetized, i.e., their Larmor radii should be less than the pinch radius. For alpha-particles produced in the D–T reaction, this condition is satisfied at pinch currents higher than 300 kA.

Second, it is necessary that the diffusion of alpha-particles be slow enough for their energy to be released in the adjacent plasma regions. Unfortunately, the diffusion rate of alpha-particles has been poorly studied yet. It is only known that, if the diffusion of alpha-particles in the transverse magnetic field of a pinch were classical, then the energy of the charged thermonuclear products would be released locally.

It is expected that the initiation of a thermonuclear burn wave will be achieved in a deuterium–tritium Z-pinch. However, the use of tritium is forbidden in most laboratories; therefore, experiments are usually performed with deuterium (without tritium) and neutrons in them are produced the D–D reaction. In this case, the possibility of igniting a thermonuclear reaction in a D–T mixture can be estimated from the following criteria for deuterium [95, 96]:

$$Y_{\text{DD}} > 1.67 \times 10^{13} h I^2, \quad (4.4)$$

$$dY_{\text{DD}}/dt > 1.3 \times 10^{23} I^2, \quad (4.5)$$

where  $Y_{\text{DD}}$  is the neutron yield of the D–D reaction,  $h$  is the length of the area emitting neutrons (in cm), and  $I$  is the pinch current (in MA).

Conditions (4.4) and (4.5) were derived by multiplying conditions (4.2) and (4.3) by the ratio between the intensities of neutron emission in the D–D and D–T reactions (here, this ratio is taken to be 1/80).

If, in some facility, requirements (4.4) and (4.5) are satisfied with the use of a deuterium load, then requirements (4.2) and (4.3) will be automatically satisfied with the use of a D–T mixture, i.e., necessary conditions for initiating a thermonuclear burn wave will be achieved.

The neutron yield as a function of the current in gas-discharge Z-pinches was thoroughly studied experimentally. The neutron yield was found to increase with increasing current as  $Y_{\text{DD}} = 10^{11} I^4$  [18], where the discharge current  $I$  is in megaamperes. This dependence was proved to hold up to currents of 3 MA. Taking into

account that the typical length of the emitting region in those experiments was  $\sim 1$  cm, one can find from this experimental dependence that condition (4.4) for igniting a thermonuclear reaction should be satisfied at currents higher than 30 MA (Fig. 18).

Thus, the current required to ignite a thermonuclear burn wave in a Z-pinch can be estimated from the dependence of the neutron yield on the pinch current. Unfortunately, experimental data on the neutron yield at pinch currents in the range 4–20 MA are still lacking. Therefore, it is desirable to continue these investigations in order to further increase the neutron yield.

It is also very important to perform investigations aimed at increasing the neutron yield per unit length of the pinch. One way of increasing this yield is to concentrate the deposited energy in a single constriction. In this case, the higher energy density achieved over a shorter pinch length makes it easier to satisfy conditions for igniting a thermonuclear burn wave. Such investigations are currently performed in the S-300 facility at the Kurchatov Institute.

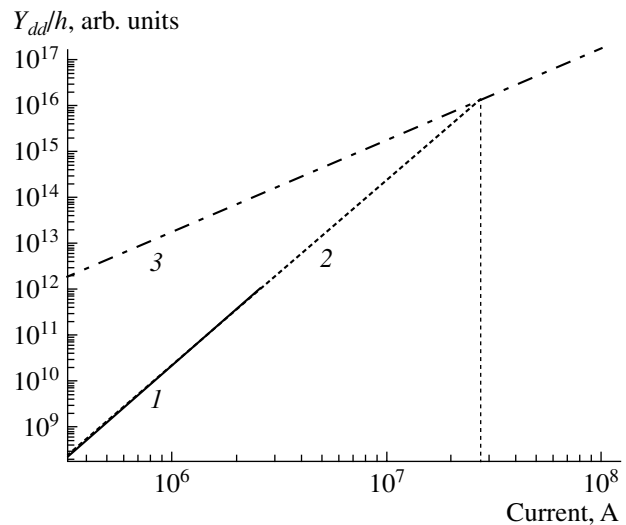
## 5. CONCLUSIONS

At present, Z-pinchs are among facilities producing record neutron yields. The maximum neutron yield ( $6 \times 10^{13}$  neutrons per shot) was achieved in experiments with a deuterium jet in the 40-TW at a discharge current of 17.6 MA. Upon upgrading the Z facility, the current in the modernized ZR facility can be increased to 30 MA. Calculations show that, in this case, the neutron yield may increase to  $2 \times 10^{15}$  neutrons per shot for a deuterium load and to  $6 \times 10^{16}$  for a deuterium–tritium load. Such a high neutron yield is a necessary condition for achieving the plasma parameters required to ignite a pulsed thermonuclear reaction.

Z-pinchs have a number of advantages over other pulsed methods for compressing and heating plasmas: the efficiency with which the energy is transferred from the generator to the load is fairly high (up to a few per cent), the energy is transferred in a small number of stages, and a thermonuclear reaction can be spark-ignited in the pinch itself.

Experimental and theoretical investigations of neutron radiation from Z-pinchs have shown that the neutron yield in these systems cannot be simply represented as the sum of the thermal and target neutrons. Neutron generation in Z-pinchs can only be described by analyzing the ion energy distribution.

The ion energy spectrum in Z-pinchs differs from a Maxwellian one: the number of fast ions is much larger than that in a Maxwellian distribution and falls off by a power law (the exponent being from  $-2$  to  $-4$ ) with increasing energy, rather than according to an exponential law as is in the case of a Maxwellian distribution. Due to such a power-law tail, the actual neutron yield from pinchs is higher than that calculated for a thermal ion distribution over energy.



**Fig. 18.** Total yield of D–D neutrons per unit length of the pinch as a function of the discharge current: (1) experiment, (2) extrapolation of the experimental data, and (3) straight line corresponding to ignition criterion (4.4).

The ion distribution in a pinch is nonequilibrium because the ions acquire energy directly from the pinch magnetic field due to the compression and turbulization of the pinch plasma, rather than from other plasma particles. Therefore, collisions between plasma particles have no time to thermalize the ion energy distribution.

If the ion energy spectrum varied self-similarly with increasing pinch energy, then the neutron-yield scaling would be proportional to the fourth power of the current. However, being deduced for a given facility, this scaling usually turns out to be weaker because the facility is not optimized over the entire range of discharge currents.

For low-power facilities, this scaling is sometimes stronger because the mean ion energy increases with discharge current. However, the dependence of the neutron yield on the pinch current is so strong only within a narrow current interval and fails to hold beyond this interval.

The neutron yield from a pinch with an energy higher than 30 MJ can be substantially increased by using a D–T mixture because of the initiation of a thermonuclear burn wave.

The anisotropy of neutron radiation stems from the directed motion of a considerable fraction of ions along the pinch axis. This is merely a consequence of the Z-pinch magnetic configuration and does not indicate a specific mechanism for ion acceleration along the pinch axis.

In order to initiate thermonuclear burn in a pinch, it is necessary that the magnetic energy density near the pinch be increased as much as possible by compressing the pinch plasma to a small radius and by maximizing the pinch current. Due to the MHD instabilities, this

magnetic energy will be transferred to the plasma particles, thereby enabling an intense thermonuclear reaction. Alpha particles generated in this region will heat the neighboring regions of the pinch plasma to thermonuclear temperatures and may thus produce a thermonuclear burn wave along the pinch. Therefore, this mechanism can be efficiently used to ignite a self-sustained thermonuclear reaction in the pinch plasma.

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